

Honors Contract: Data Analysis Project

Analyzing and Understanding Online Hate Speech Using "HateXplain" dataset

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DAT 301: Exploring Data in R and Python

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I. Introduction

This project analyzes hate speech in online comments, focusing on how different demographic groups are targeted and patterns of speech. The goal of this analysis is not only to identify which groups are most frequently targeted, but also to examine how hate speech, offensive language, and normal comments differ in their distribution. Understanding these patterns is important because online comment may reflect the concealed reality of hate toward a specific demographic groups because of its anonymity and dehumanization aspect. By analyzing hate speeches through online comment, we can better understand how digital environment reflect, reproduce, or amplify real-world discrimination.

i) Variable Definition

This analysis used "final_hateXplain" of "HateXplain" dataset from Kaggle. The dataset includes the following seven variables (columns): comment, label, race, religion, gender, sexual orientation, and miscellaneous. Each "comment" variables are assigned one of the categories of other six variables.

1. Comment: the original comments online.
2. Label: includes three categories: hate, offensive, and normal.
3. Race: includes 8 categories: no race, African, Arab, Caucasian, Asian, Hispanic, Indigenous, and Indian.
4. Gender: includes 3 categories: no gender, women, and men.
5. Religion: includes 6 categories: nonreligious, Islam, Jewish, Christian, Hindu, and Buddhism.
6. Sexual Orientation: includes 5 categories: no orientation, homosexual, heterosexual, asexual, and bisexual.

7. Miscellaneous: includes 5 categories: other, refugee, minority, disability, and economic.

II. Dataset and Methods

The analysis was conducted using Python with libraries such as pandas, matplotlib, and seaborn. The following steps were performed:

1. Dataset overview and structure analysis
2. Distribution analysis of demographic targets
3. Filtering out comments with no specific target
4. Label distribution analysis
5. Linguistic feature analysis (e.g. text length)
6. Identification of top targeted groups based on its label - hate, offensive, and normal

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import re

# Load the dataset
df = pd.read_csv('final_hateXplain.csv')
```

III. Results

1. Dataset Overview

```
In [2]: # 1. check the dataset (txt file + overview)
with open('1_dataset_info.txt', 'w') as f:
    # to screen
    print("--- Data Outline ---")
    overview = f"Rows: {df.shape[0]}, Columns: {df.shape[1]}"
    print(overview)

    # to txt file
    f.write("--- Data Overview ---\n")
    f.write(f"Dataset Shape: {df.shape}\n")
    f.write(f"Columns: {df.columns.tolist()}\n")

    f.write("\n --- variable info ---\n")
    var_info = pd.DataFrame({
        "Column": df.columns,
        "Dtype": df.dtypes.values
    })
    print(var_info)
    f.write(var_info.to_string() + "\n\n")

    # check for missing values
    print("\n --- missing values ---")
```

```

f.write("\n --- missing value count ---\n")

null_counts = df.isnull().sum()
print(null_counts)
f.write(null_counts.to_string() + "\n\n")

# check missing value percentage
null_pct = (df.isnull().sum() / len(df)) * 100
print("\n --- missing value percentage ---")
f.write("\n --- missing value percentage ---\n")
print(null_pct)
f.write(null_pct.to_string() + "\n\n")

print('--- variable info recorded in 1_dataset_info.txt ---')

```

```

--- Data Outline ---
Rows: 20109, Columns: 7

```

	Column	Dtype
0	comment	object
1	label	object
2	Race	object
3	Religion	object
4	Gender	object
5	Sexual Orientation	object
6	Miscellaneous	object

```

--- missing values ---
comment          0
label            0
Race            0
Religion        0
Gender          0
Sexual Orientation 0
Miscellaneous    16576
dtype: int64

```

```

--- missing value percentage ---
comment          0.000000
label            0.000000
Race            0.000000
Religion        0.000000
Gender          0.000000
Sexual Orientation 0.000000
Miscellaneous    82.430752
dtype: float64

```

```

--- variable info recorded in 1_dataset_info.txt ---

```

The dataset contains a total of 20109 number of rows with 7 columns. Missing value was checked, and there were 16576 missig values (82.43%) in Miscellaneous column. Checking missing values is important for more precise analysis.

2. Demographic Target Distribution Analysis

```

In [3]: # 2. Demographic (target) distribution anlysis
        ## check each demographic (target) variable
        target_cols = ['Race', 'Religion', 'Gender', 'Sexual Orientation', 'Miscellaneous']

        ## record to txt file

```

```

with open('2_target_variable_check.txt', 'w') as f:
    f.write("--- Target Columns Variable Check ---\n\n")

    for col in target_cols:
        print(f'variable check: {col.upper()}')
        ### print all values and counts in each column
        target_counts = df[col].value_counts()

        print(target_counts)
        print("-" * 30)

        ### write to txt file
        f.write(f'[{col.upper()}]\n')
        f.write(target_counts.to_string() + "\n\n")

print("\n --- Target variable check recorded in 2_target_variable_check.txt ---")

non_values = {'Race': 'No_race', 'Religion': 'Nonreligious', 'Gender': 'No_gender',
              'Sexual Orientation': 'No_orientation'}

### pick the rows with no specific target
is_none = df.apply(
    lambda x: all(x[col] == val for col, val in non_values.items()),
    axis=1,
)
non_count = is_none.sum()
non_df = df[is_none]
non_label_counts = non_df['label'].value_counts()
non_label_pct = non_df['label'].value_counts(normalize=True) * 100

print(f"Among total {len(df)} rows, the rows with no specific target for all lab")
print(f"Ratio: {(non_count/len(df)) * 100:.2f}")
print("\n--- Label distribution among all no specific target rows ---")
print(non_label_counts)

with open('2_non_value_analysis.txt', 'w') as f:
    f.write("--- Analysis of rows with all no specific target values ---\n\n")
    f.write(f"Total rows: {len(df)}\n")
    f.write(f"Rows with all no specific target values: {non_count}\n")
    f.write(f"Ratio: {(non_count/len(df)) * 100:.2f}%\n")
    f.write("\n--- Label Distribution ---\n")
    f.write(non_label_counts.to_string() + "\n\n")
    f.write("--- Label Distribution Percentage ---\n")
    for label, pct in non_label_pct.items():
        f.write(f"{label}: {pct:.2f}%\n")

non_summary_table = pd.DataFrame({
    'Label': non_label_counts.index,
    'Count': non_label_counts.values,
    'Percentage': [f'{pct:.2f}%' for pct in non_label_pct.values]
})

fig, ax = plt.subplots(figsize=(8, max(4, len(non_summary_table) * 0.8 + 2)))
ax.axis('off')

summary_text = (
    f'Total rows: {len(df)}\n'
    f'All no(n)_ rows: {non_count}\n'
    f'Ratio: {(non_count/len(df)) * 100:.2f}%'

```

```

)
ax.text(
    0.02,
    0.98,
    summary_text,
    transform=ax.transAxes,
    fontsize=11,
    fontweight='bold',
    va='top',
)

non_table = ax.table(
    cellText=non_summary_table.values,
    colLabels=non_summary_table.columns,
    loc='center',
    colLoc='center',
    cellLoc='center',
    bbox=[0, 0.05, 1, 0.65],
)
non_table.auto_set_font_size(False)
non_table.set_fontsize(10)

for (row, col_idx), cell in non_table.get_celld().items():
    if row == 0:
        cell.set_text_props(fontweight='bold')

plt.title(
    "All No-Specific-Target Rows Analysis",
    fontsize=16,
    fontweight='bold',
    pad=20
)
plt.tight_layout()
plt.savefig('2_non_value_analysis.png', bbox_inches='tight', dpi=300)
plt.close()

print("\n --- Analysis of rows with all 'no(n)_' values recorded in 3_non_value_

## bar plot by label
lab_color = {
    'hatespeech': 'red',
    'offensive': 'orange',
    'normal': 'green'
}
plt.figure(figsize=(8, 8))

sns.barplot(
    x=non_label_counts.index,
    y=non_label_counts.values,
    palette=lab_color,
    legend=False
)

plt.title('Label Distribution among No Specific Target Comments', fontsize=16, f
plt.xlabel('Label', fontsize=12, fontweight='bold')
plt.ylabel('Count', fontsize=12, fontweight='bold')
plt.tight_layout()
plt.savefig('2_non_label_distribution.png', bbox_inches='tight', dpi=300)
plt.show()

```

```

variable check: RACE
Race
No_race      13579
African      3769
Arab         1172
Caucasian    802
Asian        400
Hispanic     313
Indigenous   42
Indian       32
Name: count, dtype: int64

```

```

-----
variable check: RELIGION
Religion
Nonreligious  15387
Islam        2559
Jewish       1950
Christian    163
Hindu        42
Buddhism     8
Name: count, dtype: int64

```

```

-----
variable check: GENDER
Gender
No_gender    16380
Women        2262
Men          1467
Name: count, dtype: int64

```

```

-----
variable check: SEXUAL ORIENTATION
Sexual Orientation
No_orientation  17827
Homosexual     2163
Heterosexual   113
Asexual        4
Bisexual       2
Name: count, dtype: int64

```

```

-----
variable check: MISCELLANEOUS
Miscellaneous
Other         2017
Refugee       1274
Minority      107
Disability    96
Economic      39
Name: count, dtype: int64

```

--- Target variable check recorded in 2_target_variable_check.txt ---

Among total 20109 rows, the rows with no specific target for all labels: 7629
Ratio: 37.94

--- Label distribution among all no specific target rows ---

```

label
normal      5873
offensive   1506
hatespeech  250
Name: count, dtype: int64

```

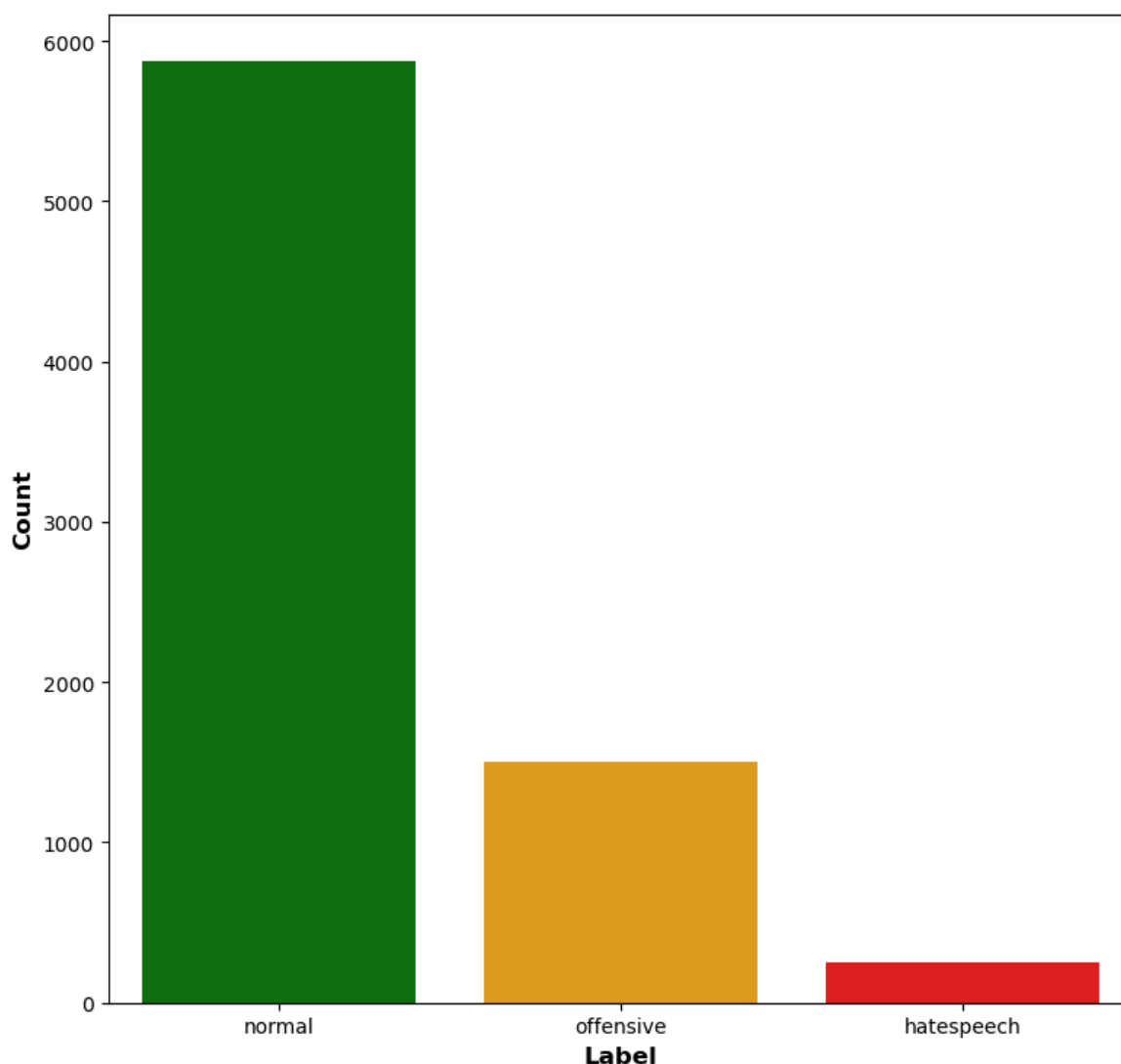
```
--- Analysis of rows with all 'no(n)_' values recorded in 3_non_value_analysis.txt ---
```

```
C:\Users\lsm11\AppData\Local\Temp\ipykernel_14188\1693052830.py:112: FutureWarning:
```

```
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.
```

```
sns.barplot(
```

Label Distribution among No Specific Target Comments



First, demographic analysis was conducted: The number of each categories in race, religion, gender, sexual orientation, and miscellaneous variables was analyzed. A large portion of comments does not target any specific group. However, among targeted comments, race-related categories appear most frequently, followed by religion, gender, and sexual orientation. Within these categories, certain groups are targeted more often than others. For example, African and Arab are the most frequently targeted racial groups, while Islam and Jewish are the most frequently targeted religions groups. This indicates that targeting is not evenly distributed across groups.

Second, the distribution of comments with no specific target was visualized. Because most of the comments appears not to target a specific demographic group, separate analysis for targetted and non-targetted comment is needed to better understand the hate speech trend. Each labels are colored differently: red color for hate speech, orange color for offensive speech, and green color for normal speech. The distribution shows that the frequency of normal speech is significantly higher than the offensive and hate speech.

3. Target Distribution Excluding No Specific Groups

```
In [4]: # 3. check each target distribution except for "no(n)_" values
### filter no(n) value for each target
none_map = {
    'Race': 'No_race',
    'Religion': 'Nonreligious',
    'Gender': 'No_gender',
    'Sexual Orientation': 'No_orientation'
}

### plot distribution for each target column and save to file
print('--- Plotting target variable distributions (excluding "no(n)_" values) ---')

with open('3_filtered_target_distribution_analysis.txt', 'w') as f:
    for col, none_val in none_map.items():
        filtered_df = df[df[col] != none_val]
        counts = filtered_df[col].value_counts()

        print(f'\n>>> {col.upper()} (total actual target data: {len(filtered_df)})')
        print(counts)

        ### write to txt file
        f.write(f'[{col.upper()}] Distribution (excluding "no(n)_" values)\n')
        f.write(counts.to_string() + '\n\n')

        ### visualization: bar plot using matplotlib & seaborn (for each column)
        ### save to (4_dist_{col}.png)
        if not counts.empty:
            plt.figure(figsize=(10, 6))
            sns.barplot(
                x=counts.index,
                y=counts.values,
                hue=counts.index,
                palette='coolwarm',
                legend=False,
            )
            plt.title(f'Actual Targets in {col.capitalize()} (excluding {none_val})')
            plt.xlabel('Group')
            plt.ylabel('Count')
            plt.xticks(rotation=45)
            plt.tight_layout()
            plt.savefig(f'4_filtered_dist_{col}.png')
            plt.show()

misc_counts = df['Miscellaneous'].value_counts()
```

```

print(misc_counts)

with open('3_filtered_target_distribution_analysis.txt', 'a') as f:
    f.write('[MISCELLANEOUS] Distribution\n')
    f.write(misc_counts.to_string() + '\n\n')

plt.figure(figsize=(10, 6))
sns.barplot(
    x=misc_counts.index,
    y=misc_counts.values,
    hue=misc_counts.index,
    palette='coolwarm',
    legend=False,
)
plt.title('Actual Targets in Miscellaneous (including Other)')
plt.xlabel('Group')
plt.ylabel('Count')
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig('3_filtered_dist_Miscellaneous.png')
plt.show()

print('\n--- Filtered Target Distribution Summary and Barplot saved ---')

```

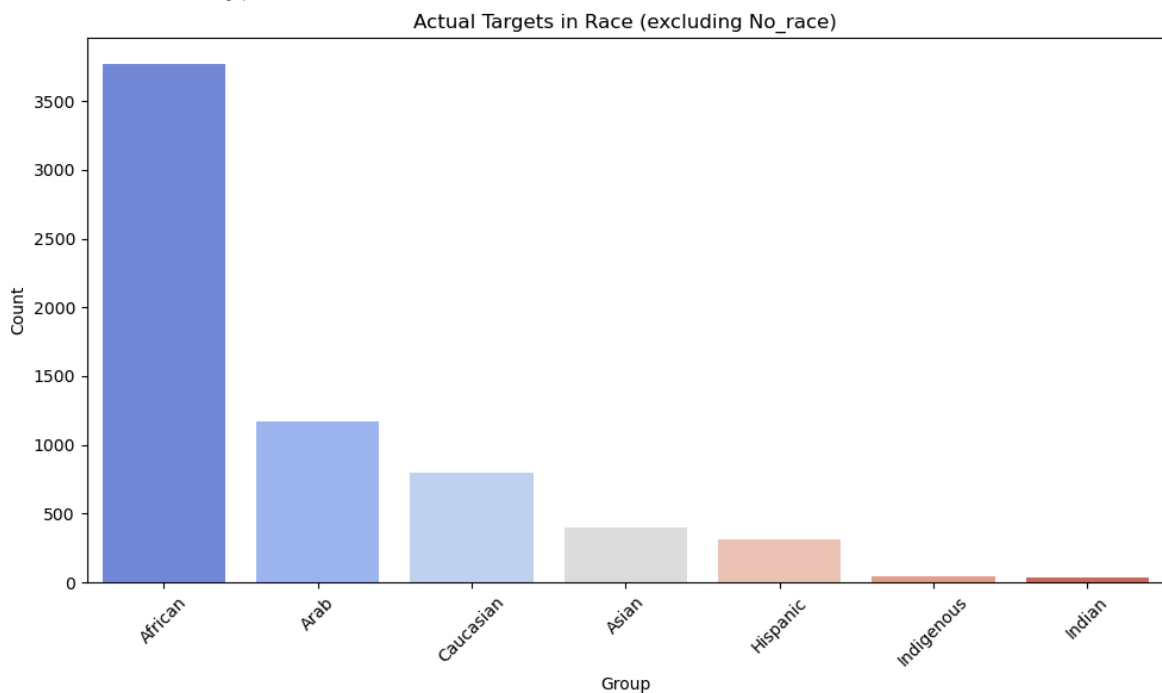
--- Plotting target variable distributions (excluding "no(n)_" values) ---

>>> RACE (total actual target data: 6530)

Race

African	3769
Arab	1172
Caucasian	802
Asian	400
Hispanic	313
Indigenous	42
Indian	32

Name: count, dtype: int64



```
>>> RELIGION (total actual target data: 4722)
```

```
Religion
```

```
Islam      2559
```

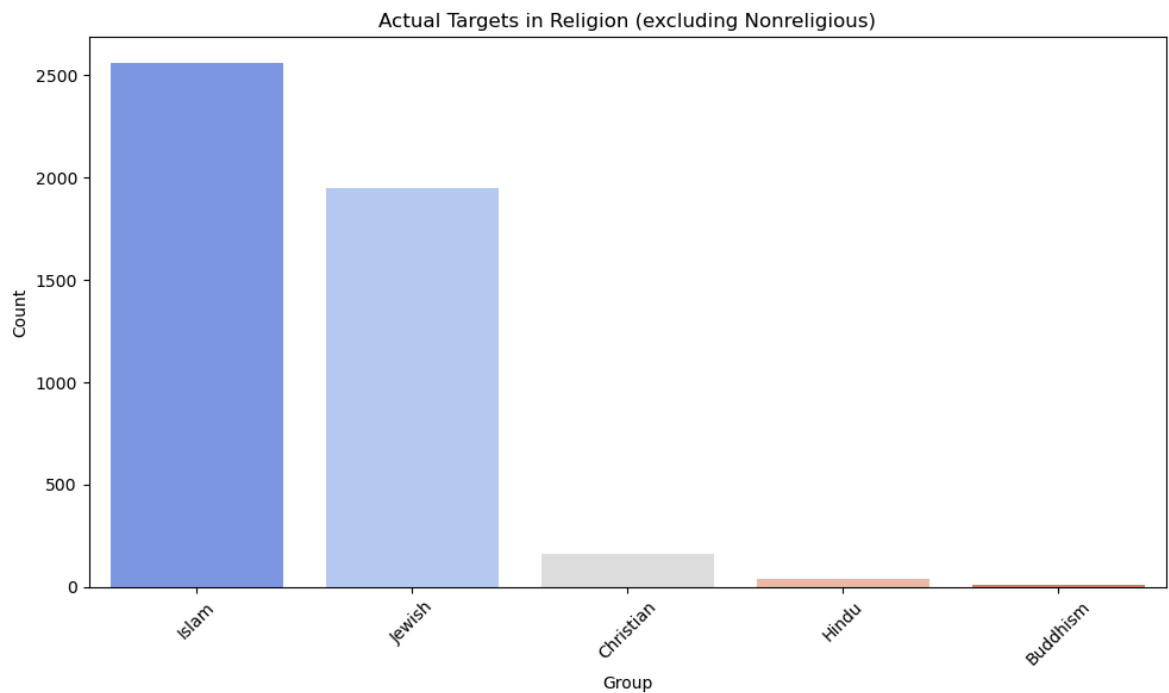
```
Jewish     1950
```

```
Christian   163
```

```
Hindu       42
```

```
Buddhism    8
```

```
Name: count, dtype: int64
```



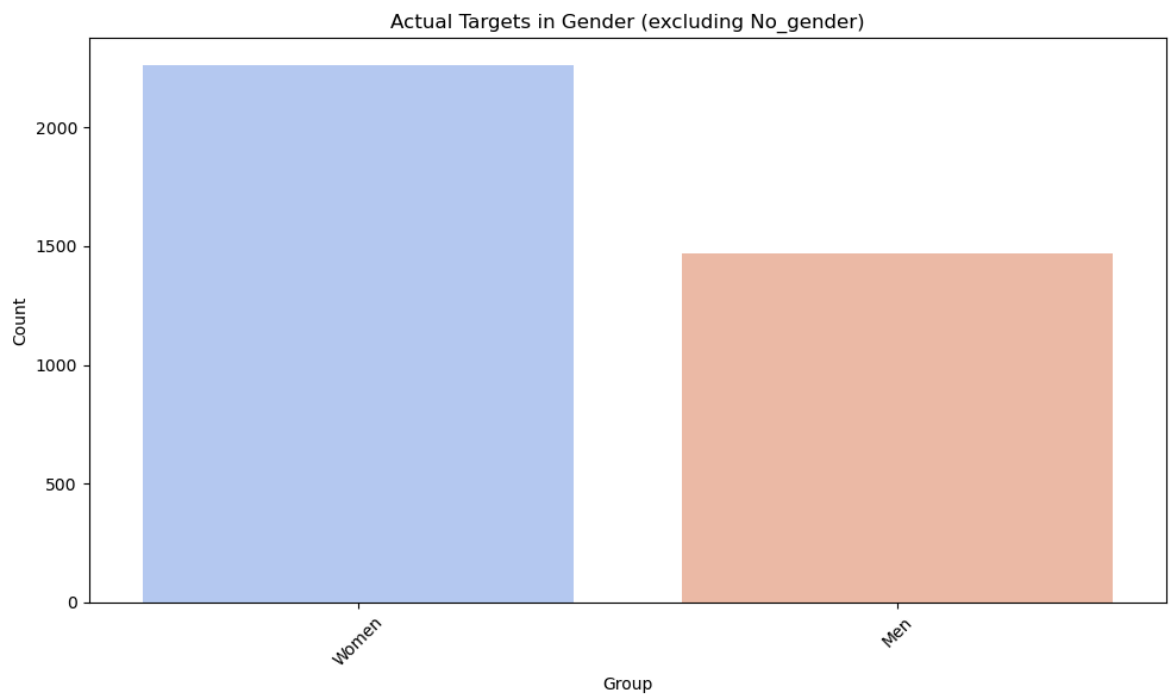
```
>>> GENDER (total actual target data: 3729)
```

```
Gender
```

```
Women     2262
```

```
Men        1467
```

```
Name: count, dtype: int64
```



```
>>> SEXUAL ORIENTATION (total actual target data: 2282)
```

```
Sexual Orientation
```

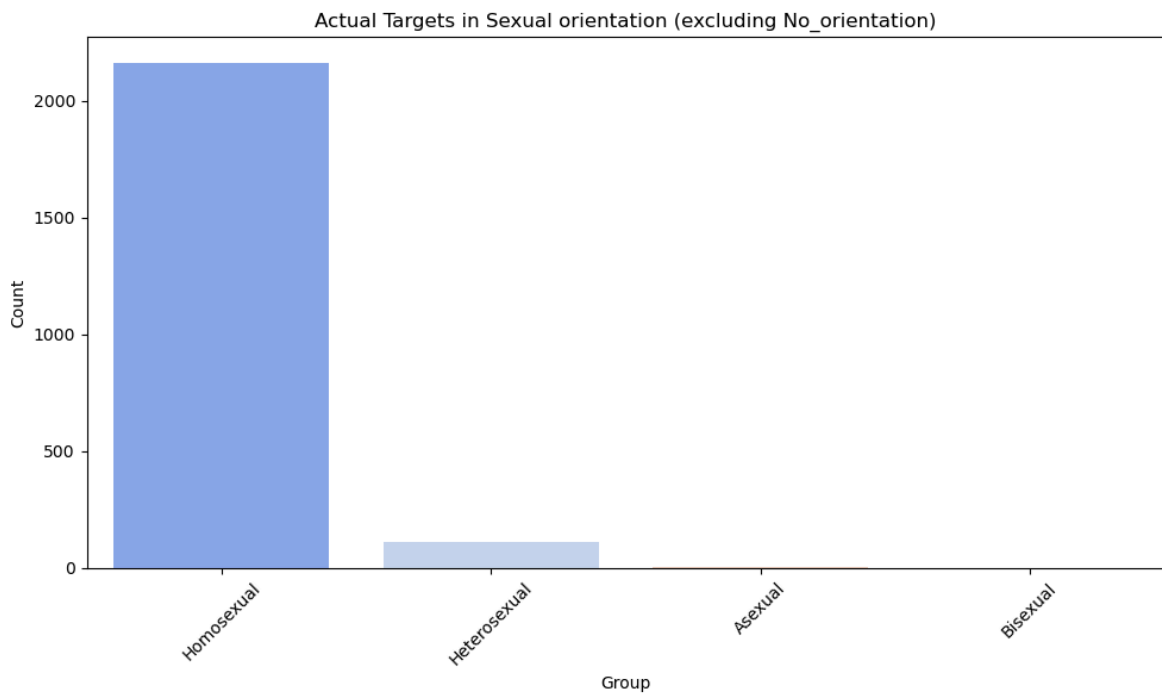
```
Homosexual      2163
```

```
Heterosexual    113
```

```
Asexual         4
```

```
Bisexual        2
```

```
Name: count, dtype: int64
```



```
Miscellaneous
```

```
Other           2017
```

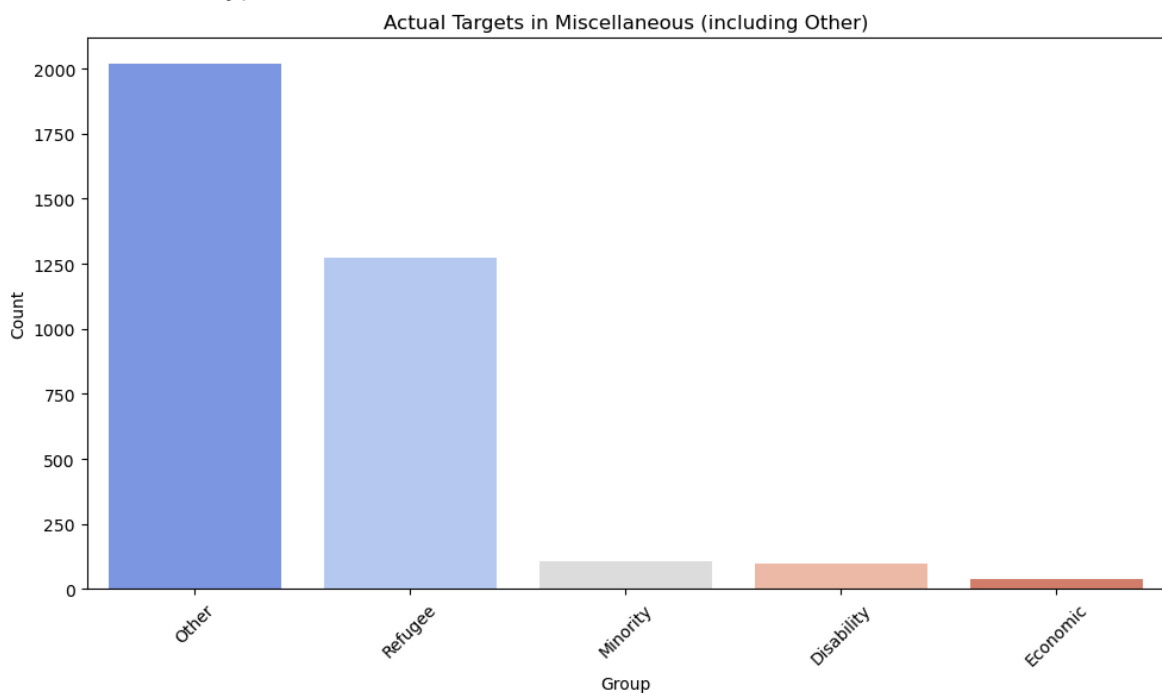
```
Refugee         1274
```

```
Minority        107
```

```
Disability       96
```

```
Economic        39
```

```
Name: count, dtype: int64
```



--- Filtered Target Distribution Summary and Barplot saved ---

This part of analysis focuses on how often a specific group was targeted, but not by its speech label. For each variables, there are quite extreme disparities in its frequency

of being targeted. Regarding race, African group are significantly most targeted group (3,769 out of 6,530). Regarding religion, Islam (2,559) and Jewish (1,950) are two most frequently targeted group among a total of 4,772. Regarding gender, women are more often targeted (2,262) than men (1,467). Regarding sexual orientation, homosexual group consists most of targeted sexual orientation (2,163 out of 2,282). For miscellaneous, other (2,017) and refugee (1,274) consists most of targeted miscellaneous categories.

4. Label Distribution Excluding No Specific Target Comments

```
In [5]: # 4. check label distribution except for no specific target comments
## check frequency
print('--- Checking label distribution excluding no specific target comments ---')

targeted_df = df[~is_none]

targeted_label_count = targeted_df['label'].value_counts()
targeted_label_pct = targeted_df['label'].value_counts(normalize=True) * 100
print(targeted_label_count)

## save to txt file
with open('4_label_distribution.txt', 'w') as f:
    f.write("--- Label Distribution ---\n\n")
    f.write(targeted_label_count.to_string() + '\n\n')
    # add ratio (%)
    targeted_label_pct = targeted_df['label'].value_counts(normalize=True) * 100
    f.write('--- Percentage Table --- \n')
    for label, pct in targeted_label_pct.items():
        f.write(f'{label}: {pct:.2f}%\n')

## visualize 1: pie chart
plt.figure(figsize=(8, 8))
### label color: Hate speech (Red), Offensive (Orange), Normal (Green)
label_color = {
    'hatespeech': 'red',
    'offensive': 'orange',
    'normal': 'green'
}

### draw bar plot & pie chart and save to file (5_label_distribution_pie.png)
plt.figure(figsize=(8, 8))
sns.barplot(
    x=targeted_label_count.index,
    y=targeted_label_count.values,
    hue=targeted_label_count.index,
    palette=label_color,
    legend=False,
)
plt.title('Label Distribution w/ No Specific Target Comments (Bar Plot)')
plt.xlabel('Label')
plt.ylabel('Count')
plt.savefig('4_label_distribution_bar.png')

plt.figure(figsize=(8, 8))
```

```
plt.pie(targeted_label_count.values, labels=targeted_label_count.index,  
        colors=[label_color[label] for label in targeted_label_count.index], autop  
plt.title('Label Distribution w/ no specific target comments (Pie Chart)')  
plt.savefig('4_label_distribution_pie.png')  
plt.show()  
  
print('--- Label Distribution Summary and Visualizations saved ---')
```

--- Checking label distribution excluding no specific target comments ---

label

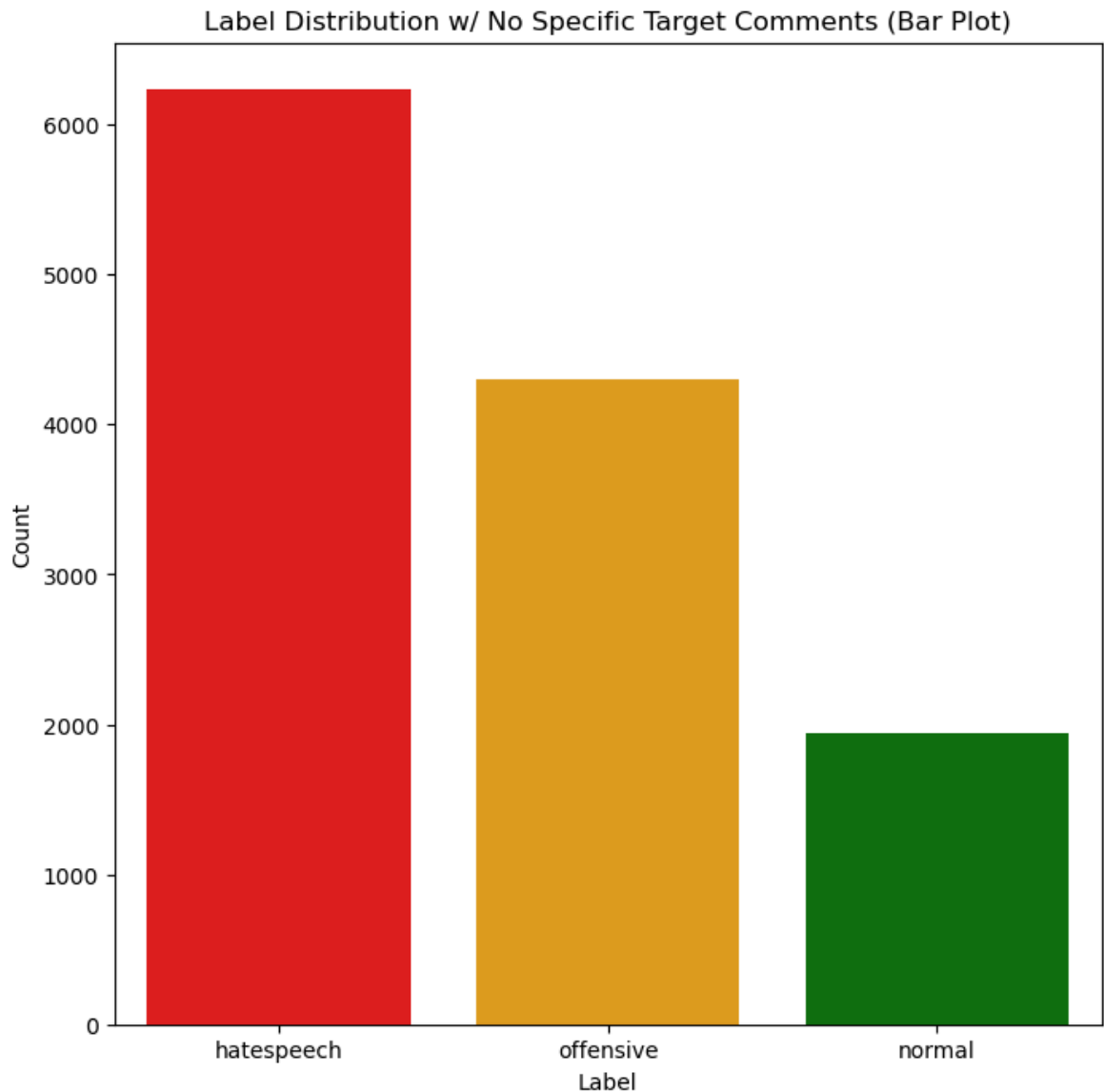
hatespeech 6234

offensive 4301

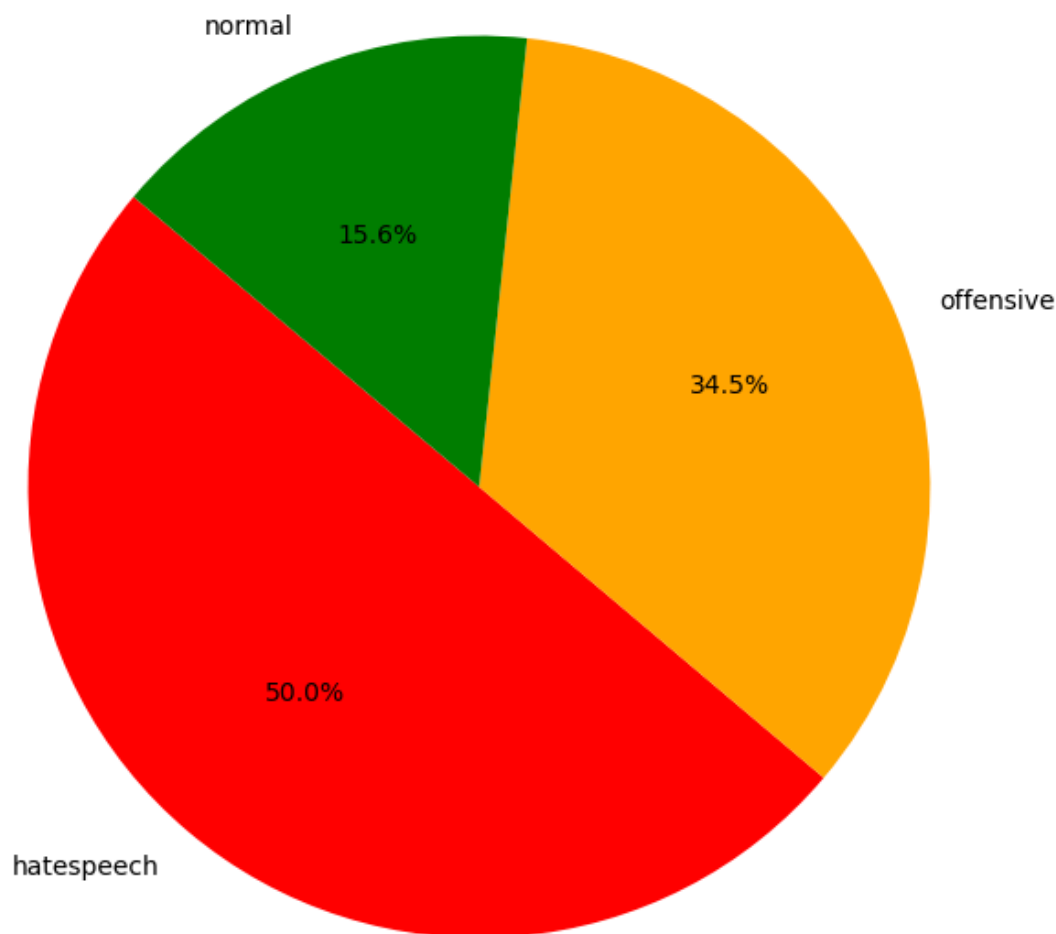
normal 1945

Name: count, dtype: int64

<Figure size 800x800 with 0 Axes>



Label Distribution w/ no specific target comments (Pie Chart)



--- Label Distribution Summary and Visualizations saved ---

When analyzing targeted comments, there are big differences in speech label distributions compared to non target comments. A bar plot and a pie chart are both used for a better comparison instead of a total count of targeted comments. When focusing only on targeted comments, hate speech consists a half of all targeted comments (6,234, 50.0%). Offensive speech is the second frequent (4,301, 34.5%), and the least frequent is normal speech (1,945, 15.6%). This result shows that when the comment targets a specific group, it is more likely to be hate or offensive speech than normal speech. In other words, it suggests that when a specific group is mentioned, the likelihood of harmful language increases.

5. Label Distribution for Each Target

```
In [6]: # 5. Label Distribution for each Target
print('--- Checking label distribution for each target ---')

## 'no(n)_' values included
target_cols = ['Race', 'Religion', 'Gender', 'Sexual Orientation', 'Miscellaneous']
```

```

print("\n>>> 100% stacked bar chart for each target (including 'no(n)_' values)"

with open('5_label_distribution_ratio_tables.txt', 'w') as f:
    f.write("--- Label Distribution Ratio Tables by Target ---\n\n")

    for col in target_cols:
        ### cross-tabulation of label and target column
        counts = pd.crosstab(df[col], df['label'])

        ### calculate percentage using row-wise normalization
        pct = pd.crosstab(df[col], df['label'], normalize='index') * 100
        sort_labels = [label for label in ['hatespeech', 'offensive', 'normal']]
        if sort_labels:
            counts = counts.sort_values(by=sort_labels, ascending=True)
            pct = pct.loc[counts.index]

        pct_formatted = pct.copy()
        for label in pct_formatted.columns:
            pct_formatted[label] = pct_formatted[label].map(lambda value: f'{val

f.write(f'[{col.upper()}]\n')
f.write(pct_formatted.to_string())
f.write('\n\n')

print(f'{col.capitalize()} ratio table image')

fig, ax = plt.subplots(figsize=(10, max(4, len(pct_formatted) * 0.6 + 1.
ax.axis('off')
ratio_table = ax.table(
    cellText=pct_formatted.values,
    colLabels=pct_formatted.columns,
    rowLabels=pct_formatted.index,
    loc='center',
    colLoc='center',
)
ratio_table.auto_set_font_size(False)
ratio_table.set_fontsize(10)

for (row, col_idx), cell in ratio_table.get_celld().items():
    if row == 0 or col_idx == -1:
        cell.set_text_props(fontweight='bold')

plt.title(
    f'Label Distribution Ratio Table: {col}',
    fontsize=16,
    fontweight='bold',
    pad=20,
)
plt.tight_layout()
plt.savefig(f'5_RatioTable_{col}.png', bbox_inches='tight', dpi=300)
plt.close()

print(f'{col.capitalize()} stacked bar chart')

plot_order = ['hatespeech', 'offensive', 'normal']
plot_cols = [label for label in plot_order if label in pct.columns]
pct_plot = pct[plot_cols]

color_map = {
    'hatespeech': 'red',

```

```

        'offensive': 'orange',
        'normal': 'green',
    }

    fig, ax = plt.subplots(figsize=(14, max(6, len(pct_plot) * 0.8)))
    pct_plot.plot(
        kind='bar',
        stacked=True,
        color=[color_map[label] for label in plot_cols],
        ax=ax,
        width=0.8,
    )

    for container, label in zip(ax.containers, plot_cols):
        labels = []
        for value in pct_plot[label]:
            labels.append(f'{value:.1f}%' if value >= 5 else '')
        ax.bar_label(container, labels=labels, label_type='center', fontsize=10)

    totals = counts.sum(axis=1)
    tick_labels = [f'{group}\n(n={totals[group]})' for group in pct_plot.index]
    ax.set_xticks(range(len(pct_plot.index)))
    ax.set_xticklabels(tick_labels, rotation=45, ha='right')
    ax.set_ylim(0, 100)
    ax.set_ylabel('Percentage (%)', fontsize=12, fontweight='bold')
    ax.set_xlabel(col, fontsize=12, fontweight='bold')
    ax.set_title(
        f'Label Distribution by {col} Sub-groups',
        fontsize=16,
        fontweight='bold',
        pad=20,
    )
    ax.legend(title='Label')
    plt.tight_layout()
    plt.savefig(f'5_StackedBar_{col}.png', bbox_inches='tight', dpi=300)
    plt.show()

    print(f'--- Label distribution ratio table saved as 5_RatioTable_{col}.p')
    print(f'--- Label Distribution by Target Subgroups saved as 5_StackedBar_{col}.png')

print('--- Label distribution ratio tables recorded in 5_label_distribution_ratio')

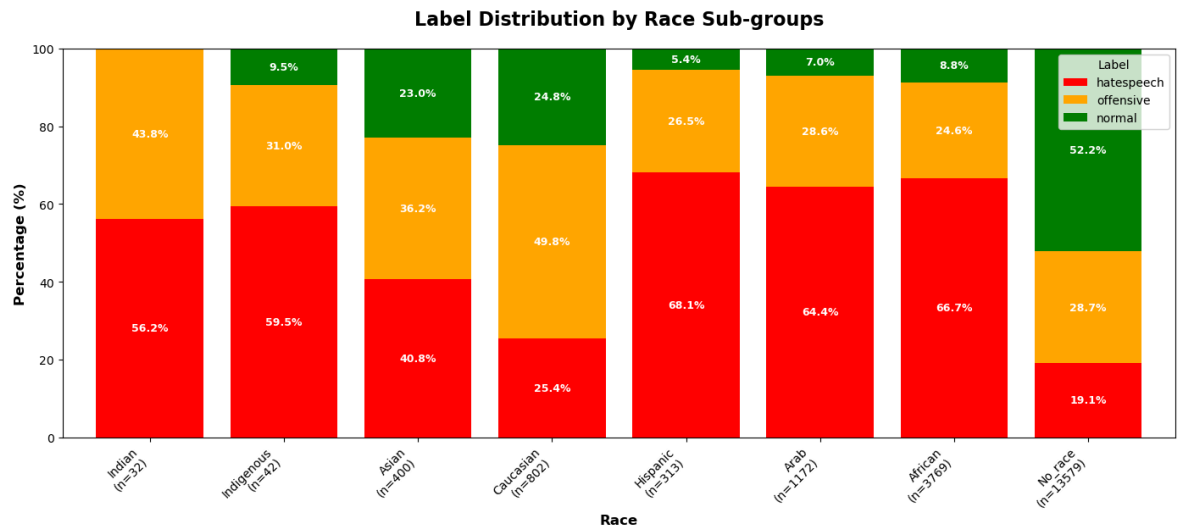
```

--- Checking label distribution for each target ---

```

>>> 100% stacked bar chart for each target (including 'no(n)_' values)
Race ratio table image
Race stacked bar chart

```

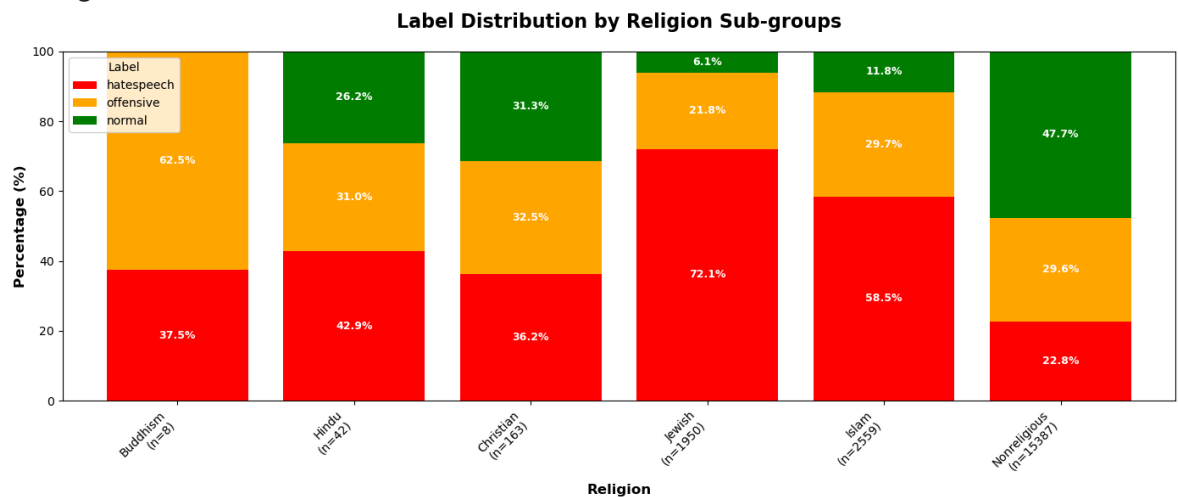


--- Label distribution ratio table saved as 5_RatioTable_Race.png ---

--- Label Distribution by Target Subgroups saved as 5_StackedBar_Race.png ---

Religion ratio table image

Religion stacked bar chart

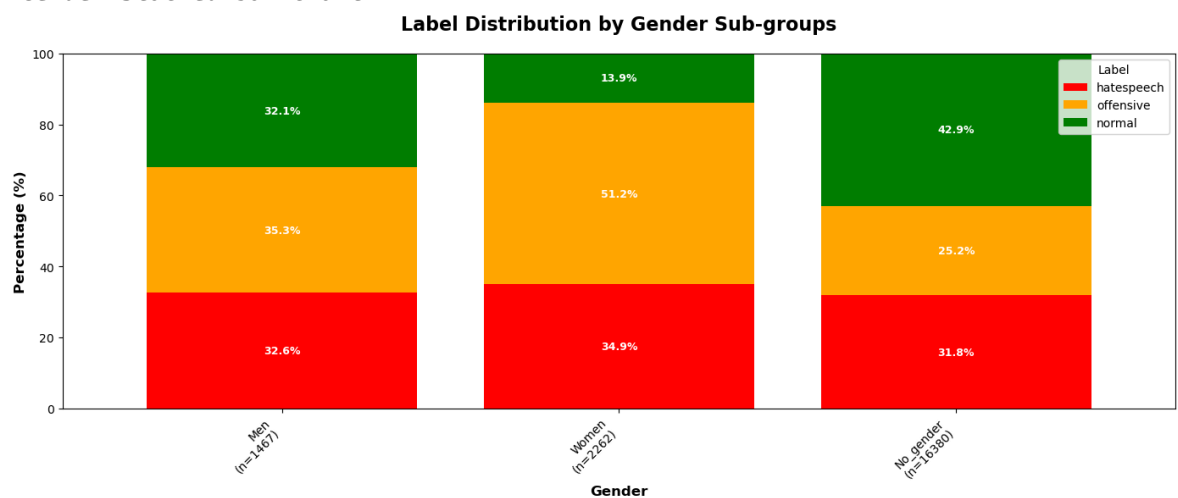


--- Label distribution ratio table saved as 5_RatioTable_Religion.png ---

--- Label Distribution by Target Subgroups saved as 5_StackedBar_Religion.png ---

Gender ratio table image

Gender stacked bar chart

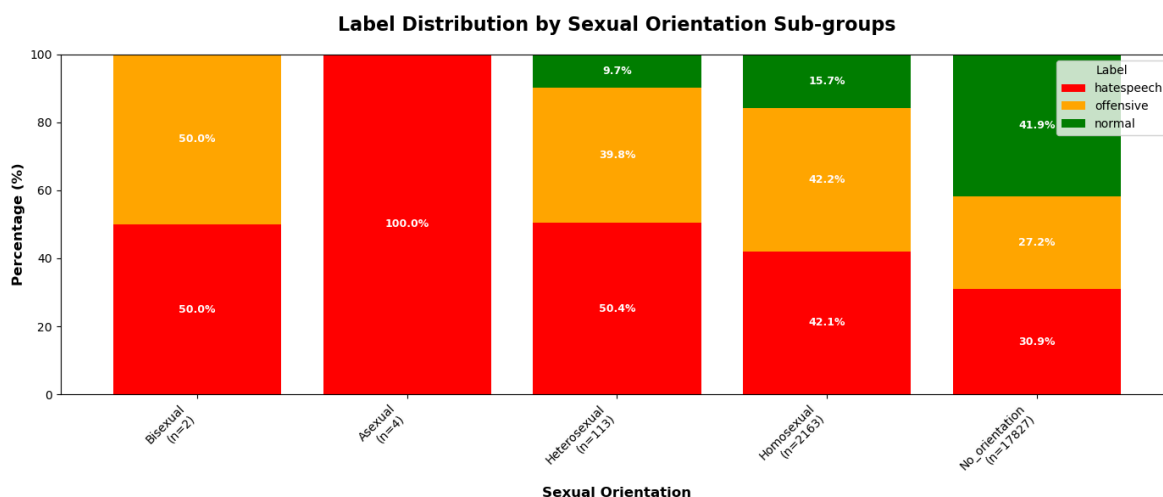


--- Label distribution ratio table saved as 5_RatioTable_Gender.png ---

--- Label Distribution by Target Subgroups saved as 5_StackedBar_Gender.png ---

Sexual orientation ratio table image

Sexual orientation stacked bar chart



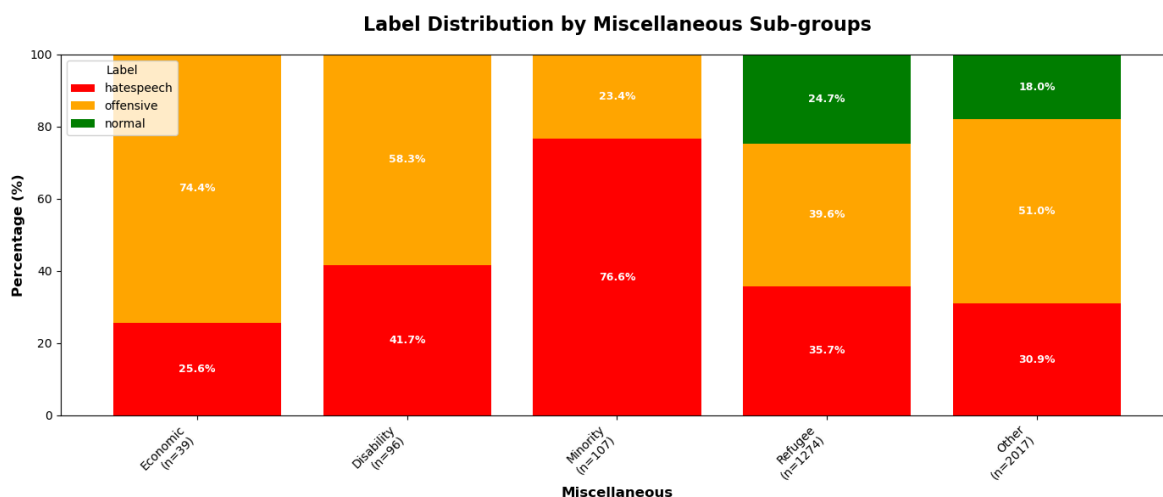
--- Label distribution ratio table saved as 5_RatioTable_Sexual Orientation.png -

--

--- Label Distribution by Target Subgroups saved as 5_StackedBar_Sexual Orientation.png ---

Miscellaneous ratio table image

Miscellaneous stacked bar chart



--- Label distribution ratio table saved as 5_RatioTable_Miscellaneous.png ---

--- Label Distribution by Target Subgroups saved as 5_StackedBar_Miscellaneous.png ---

--- Label distribution ratio tables recorded in 5_label_distribution_ratio_tables.txt ---

This part shows an a label distribution for each categories in all demographic groups. Stacked bar charts were used to visualize the frequency of each label wihtin a specific demographic groups.

When regarding race, every race group received hate speech the most, and then followed by offensive speech, except Caucasian even they are the third most targeted group. No specific race target comment showed the opposite trend. When regarding religion, the trend is quite similar to those of race. When a specific group is mentioned, hate speech is most frequent and followed by offensive speech. When regarding gender, women received offensive speech the most, while men received quite evenly distributed comment. In contrast, comments targets no specific gender, the offensive speech consists the least of total comments. When regarding miscellaneous, the speech label trends is quite different from other demographic groups. Comments about minority, disability, economic are either hate speech or

offensive, while comments about refugee and other show similar trends to the above demographic groups.

One significant pattern is that some subgroups only have either hate speech or offensive comments, not normal comments: Indian (n = 32, hate speech = 43.8%, offensive = 56.2%) in the race group, Buddhism (n = 8, hate speech = 37.5%, offensive = 62.5%) in the religion group, bisexual (n = 2, hate speech = 50.0%, offensive = 50.0%) and asexual (n = 4, hate speech = 100.0%) in sexual orientation group, along with economic (n = 39, hate speech = 25.6%, offensive = 74.5%), disability (n = 96, hate speech = 41.7%, offensive = 58.3%), and minority in miscellaneous group. In addition, these subgroups received a few comments regarding the number of other subgroups received. This can be interpreted as double-marginalization: they are mostly outside of the public's interest, but those who aware their existence mostly have negative attitude towards them.

When regarding the marginalized groups, other significant trend is that the percentage of normal comments in each subgroups: they are as low as 5.4% and as high as 26.2% (excluding non-target comments and groups considered as majority in English-speaking society - Caucasian, Christian, men, heterosexual). As previously assumed, this result indicate that when a specific group is mentioned, especially a marginalized group, the likelihood of harmful language increases.

Lastly, women were the only subgroup who received offensive comments (51.2%) more often than hate speech comments (34.9%). This may reflect that the hostility towards women is different from other subgroups. Hostility toward other marginalized group might be more direct online, but less visible offline. But for women, hostility might be more normalized through indirect or socially acceptable forms of language; or, offensive comments may show the implicit bias that has been embedded in everyday language.

6. Linguistic Features by Label

```
In [7]: # 6. check linguistic features by label
## check text length distribution by label
df['text_len'] = df['comment'].str.len()

## check emoji includedness
## True/False using unicode
def has_emoji(text):
    emoji_pattern = re.compile("[
        u"\U0001f600-\U0001f64f" # emoticons
        u"\U0001f300-\U0001f5ff" # symbols & pictographs
        u"\U0001f680-\U0001f6ff" # transport & map symbols
        u"\U0001f1e0-\U0001f1ff" # flags (iOS)
    ]+", flags=re.UNICODE)
    return bool(emoji_pattern.search(str(text)))

df['emoji_exists'] = df['comment'].apply(has_emoji)

print('--- Checking comment length and emoji distribution by label ---')
```

```

length_summary = df.groupby('label')['text_len'].agg(['count', 'mean', 'median',
emoji_counts = pd.crosstab(df['label'], df['emoji_exists'])
emoji_pct = pd.crosstab(df['label'], df['emoji_exists'], normalize='index') * 10

with open('6_linguistic_feature_analysis.txt', 'w') as f:
    f.write("--- Comment Length and Emoji Distribution by Label ---\n\n")
    f.write("[COMMENT LENGTH SUMMARY]\n")
    f.write(length_summary.to_string())
    f.write("\n\n[EMOJI COUNT]\n")
    f.write(emoji_counts.to_string())
    f.write("\n\n[EMOJI PERCENTAGE]\n")
    emoji_pct_formatted = emoji_pct.copy()
    for col in emoji_pct_formatted.columns:
        emoji_pct_formatted[col] = emoji_pct_formatted[col].map(lambda value: f'
    f.write(emoji_pct_formatted.to_string())
    f.write("\n")

print(length_summary)
print("\n--- Emoji count by label ---")
print(emoji_counts)

length_table = length_summary.reset_index()
length_table.columns = ['Label', 'Count', 'Mean', 'Median', 'Min', 'Max']
length_table[['Mean', 'Median', 'Min', 'Max']] = length_table[['Mean', 'Median',

emoji_table = emoji_counts.reset_index().copy()
emoji_table.columns = ['Label'] + ['No Emoji' if value is False else 'Emoji' for
emoji_table['Emoji %'] = emoji_pct.get(True, pd.Series(0, index=emoji_pct.index))
emoji_table['No Emoji %'] = emoji_pct.get(False, pd.Series(0, index=emoji_pct.in
emoji_table = emoji_table[['Label', 'No Emoji', 'Emoji', 'No Emoji %', 'Emoji %'

fig, axes = plt.subplots(2, 1, figsize=(12, 10))

axes[0].axis('off')
length_png_table = axes[0].table(
    cellText=length_table.values,
    colLabels=length_table.columns,
    loc='center',
    cellLoc='center',
    colLoc='center',
)
length_png_table.auto_set_font_size(False)
length_png_table.set_fontsize(10)
for (row, col_idx), cell in length_png_table.get_celld().items():
    if row == 0:
        cell.set_text_props(fontweight='bold')
axes[0].set_title('Comment Length Summary by Label', fontsize=15, fontweight='bo

axes[1].axis('off')
emoji_png_table = axes[1].table(
    cellText=emoji_table.values,
    colLabels=emoji_table.columns,
    loc='center',
    cellLoc='center',
    colLoc='center',
)
emoji_png_table.auto_set_font_size(False)
emoji_png_table.set_fontsize(10)
for (row, col_idx), cell in emoji_png_table.get_celld().items():

```

```

        if row == 0:
            cell.set_text_props(fontweight='bold')
axes[1].set_title('Emoji Presence Summary by Label', fontsize=15, fontweight='bo

plt.tight_layout()
plt.savefig('6_linguistic_feature_analysis.png', bbox_inches='tight', dpi=300)
plt.close()

plt.figure(figsize=(10, 6))
sns.boxplot(x='label', y='text_len', data=df, hue='label', palette=label_color,
plt.title('Comment Length Distribution by Label', fontsize=16, fontweight='bold')
plt.xlabel('Label', fontsize=12, fontweight='bold')
plt.ylabel('Comment Length', fontsize=12, fontweight='bold')
plt.tight_layout()
plt.savefig('6_comment_length_by_label.png', bbox_inches='tight', dpi=300)
plt.show()

emoji_plot = emoji_pct.copy()
emoji_plot.columns = ['No Emoji' if value is False else 'Emoji' for value in emc
emoji_plot = emoji_plot.reindex(columns=['No Emoji', 'Emoji'], fill_value=0)

fig, ax = plt.subplots(figsize=(10, 6))
emoji_plot.plot(
    kind='bar',
    stacked=True,
    color=['steelblue', 'goldenrod'],
    ax=ax,
    width=0.8,
)

for container in ax.containers:
    labels = []
    for value in container.datavalues:
        labels.append(f'{value:.1f}%' if value >= 5 else '')
    ax.bar_label(container, labels=labels, label_type='center', fontsize=9, colo

ax.set_ylim(0, 100)
ax.set_title('Emoji Presence Distribution by Label', fontsize=16, fontweight='bo
ax.set_xlabel('Label', fontsize=12, fontweight='bold')
ax.set_ylabel('Percentage (%)', fontsize=12, fontweight='bold')
ax.legend(title='Emoji')
plt.tight_layout()
plt.savefig('6_emoji_distribution_by_label.png', bbox_inches='tight', dpi=300)
plt.show()

print('--- Linguistic feature analysis recorded in 6_linguistic_feature_analysis
print('--- Linguistic feature summary table saved as 6_linguistic_feature_analys

```

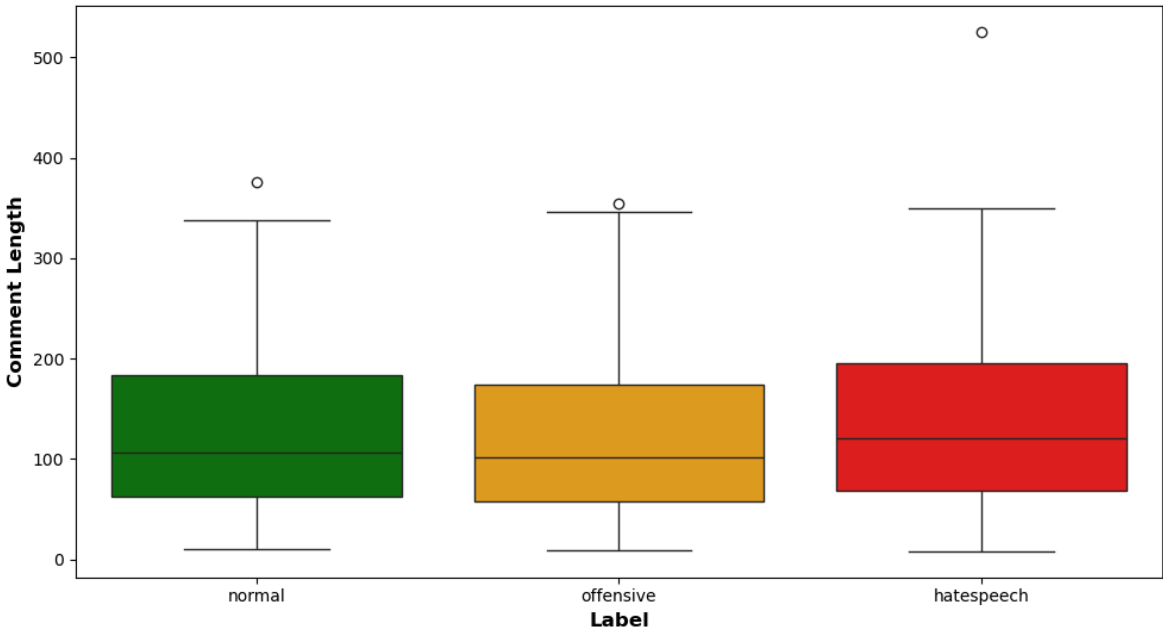
--- Checking comment length and emoji distribution by label ---

	count	mean	median	min	max
label					
hatespeech	6484	134.71	121.0	8	526
normal	7818	126.20	107.0	10	376
offensive	5807	120.87	102.0	9	354

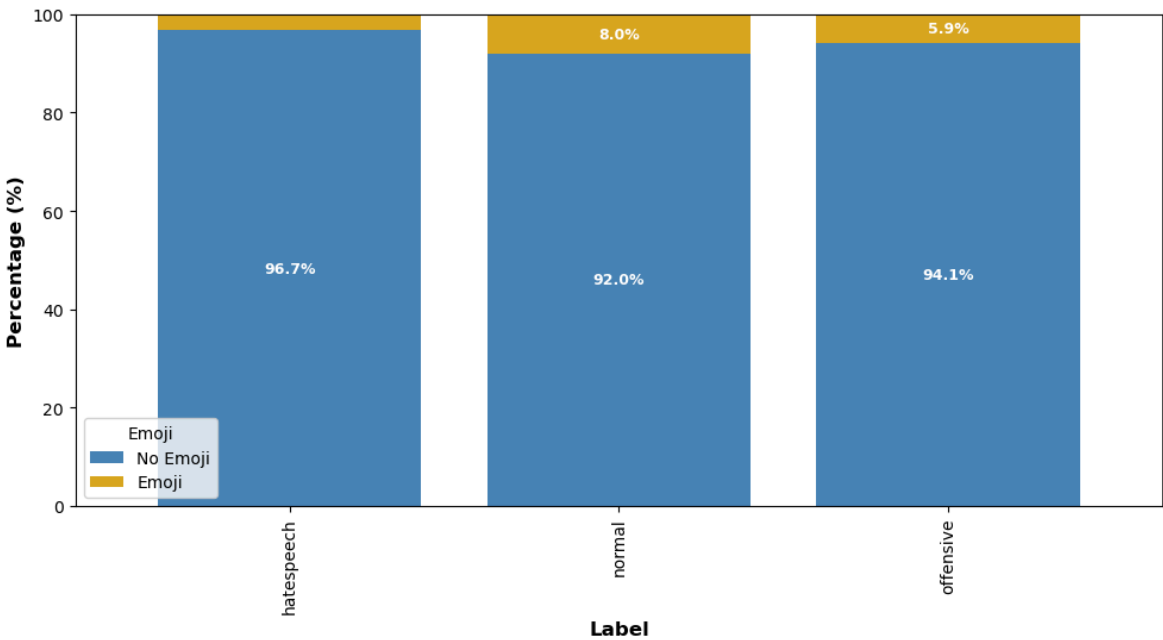
--- Emoji count by label ---

emoji_exists	False	True
label		
hatespeech	6269	215
normal	7196	622
offensive	5465	342

Comment Length Distribution by Label



Emoji Presence Distribution by Label



--- Linguistic feature analysis recorded in 6_linguistic_feature_analysis.txt ---

--- Linguistic feature summary table saved as 6_linguistic_feature_analysis.png -

--

The mean comment length and the frequency of emoji presence were analyzed to investigate if there are differences in linguistic features according to speech labels.

The mean comment length for each label was analyzed and visualized through a box plot. The box plot shows a comment length distribution by label. While the mean and the median comment length of hate speech are slightly longer(mean = 134.71, median = 121.0) compared to offensive (mean = 120.87, median = 102.0) and normal (mean = 126.2, median = 107.0) comments, the difference is relatively small, and the distributions overlap substantially. This suggest that comment length is not a strong distinguishing feature between speech labels.

Emoji usage differs across labels, with normal comments showing the highest proportion (8.0%), followed by offensive (5.9%), and hate speech (3.3%). However, emoji usage is generally low across all labels, indicating that it is not a dominant feature in distinguishing comment types.

7. Top 5 Targets with Highest Hatespeech Ratio

```
In [8]: # 7. Top 5 targets by count and ratio
analysis_df = df.copy()

# treat "no target" style values as missing targets for ranking
no_target_map = {
    'Race': 'No_race',
    'Religion': 'Nonreligious',
    'Gender': 'No_gender',
    'Sexual Orientation': 'No_orientation',
}

main_target_cols = ['Race', 'Religion', 'Gender', 'Sexual Orientation']
target_rows = []

for col in main_target_cols:
    valid_mask = analysis_df[col].notna() & (analysis_df[col] != no_target_map[col])
    col_df = analysis_df.loc[valid_mask, [col, 'label']].copy()

    for target_name, group in col_df.groupby(col):
        total_count = len(group)
        hatespeech_count = (group['label'] == 'hatespeech').sum()
        hatespeech_ratio = (hatespeech_count / total_count) * 100

        target_rows.append({
            'Target Type': col,
            'Target': target_name,
            'Total Count': total_count,
            'Hatespeech Count': hatespeech_count,
            'Hatespeech Ratio (%)': hatespeech_ratio,
        })

target_df = pd.DataFrame(target_rows)

# compare thresholds first, but use 100 for the final rankings
threshold_options = [30, 50, 100]
```

```

with open('7_top5_hatespeech_targets_comparison.txt', 'w') as f:
    f.write('--- Threshold Comparison for Ratio-Based Top 5 (excluding Miscellan

    for min_count in threshold_options:
        threshold_df = target_df[target_df['Total Count'] >= min_count].copy()
        ratio_top5 = threshold_df.sort_values(
            by=['Hatespeech Ratio (%)', 'Hatespeech Count', 'Total Count'],
            ascending=[False, False, False],
        ).head(5)

        print(f'--- Ratio Top 5 (min_count = {min_count}) ---')
        print(ratio_top5.to_string(index=False))
        print()

        f.write(f'--- min_count = {min_count} ---\n')
        f.write(ratio_top5.to_string(index=False))
        f.write('\n\n')

recommended_min_count = 100
recommended_df = target_df[target_df['Total Count'] >= recommended_min_count].co

top5_by_count = recommended_df.sort_values(
    by=['Hatespeech Count', 'Hatespeech Ratio (%)', 'Total Count'],
    ascending=[False, False, False],
).head(5)

top5_by_ratio = recommended_df.sort_values(
    by=['Hatespeech Ratio (%)', 'Hatespeech Count', 'Total Count'],
    ascending=[False, False, False],
).head(5)

print(f'Recommended minimum count threshold: {recommended_min_count}')
print('Reason: if we want groups that are actually attacked often, a larger tota
print('\n--- Top 5 targets by hatespeech count ---')
print(top5_by_count.to_string(index=False))
print('\n--- Top 5 targets by hatespeech ratio ---')
print(top5_by_ratio.to_string(index=False))

with open('7_top5_hatespeech_targets.txt', 'w') as f:
    f.write('--- Final Top 5 Targets (excluding Miscellaneous) ---\n')
    f.write(f'Minimum total count threshold: {recommended_min_count}\n')
    f.write('Reason: use 100+ observations so the top 5 reflects groups with mea
    f.write('--- Top 5 by Hatespeech Count ---\n')
    f.write(top5_by_count.to_string(index=False))
    f.write('\n\n--- Top 5 by Hatespeech Ratio (%) ---\n')
    f.write(top5_by_ratio.to_string(index=False))
    f.write('\n\n--- Full Target Table at Recommended Threshold ---\n')
    f.write(recommended_df.sort_values(
        by=['Hatespeech Count', 'Hatespeech Ratio (%)', 'Total Count'],
        ascending=[False, False, False],
    ).to_string(index=False))

fig, axes = plt.subplots(1, 2, figsize=(16, 6))

sns.barplot(
    data=top5_by_count,
    x='Hatespeech Count',
    y='Target',
    hue='Target Type',
    dodge=False,

```

```

    palette='Oranges',
    ax=axes[0],
)
axes[0].set_title(f'Top 5 Targets by Hatespeech Count (min_count = {recommended_
axes[0].set_xlabel('Hatespeech Count')
axes[0].set_ylabel('Target')
axes[0].legend(title='Target Type', loc='lower right')

sns.barplot(
    data=top5_by_ratio,
    x='Hatespeech Ratio (%)',
    y='Target',
    hue='Target Type',
    dodge=False,
    palette='Reds_r',
    ax=axes[1],
)
axes[1].set_title(f'Top 5 Targets by Hatespeech Ratio (min_count = {recommended_
axes[1].set_xlabel('Hatespeech Ratio (%)')
axes[1].set_ylabel('Target')
axes[1].legend(title='Target Type', loc='lower right')

plt.tight_layout()
plt.savefig('7_top5_hatespeech_targets.png', bbox_inches='tight', dpi=300)
plt.show()

# optional side analysis: Miscellaneous only
misc_df = analysis_df[analysis_df['Miscellaneous']].notna() & (analysis_df['Misce

if not misc_df.empty:
    misc_summary = misc_df.groupby('Miscellaneous')['label'].value_counts().unst
    misc_summary['Total Count'] = misc_summary.sum(axis=1)
    misc_summary['Hatespeech Count'] = misc_summary.get('hatespeech', 0)
    misc_summary['Hatespeech Ratio (%)'] = (
        misc_summary['Hatespeech Count'] / misc_summary['Total Count'] * 100
    )
    misc_summary = misc_summary[misc_summary['Total Count'] >= recommended_min_c
    by=['Hatespeech Count', 'Hatespeech Ratio (%)', 'Total Count'],
    ascending=[False, False, False],
)

print('\n--- Miscellaneous targets summary (separate) ---')
print(misc_summary[['Total Count', 'Hatespeech Count', 'Hatespeech Ratio (%)

    with open('7_miscellaneous_hatespeech_targets.txt', 'w') as f:
        f.write('--- Miscellaneous Target Summary ---\n')
        f.write(f'Minimum total count threshold: {recommended_min_count}\n\n')
        f.write(misc_summary[['Total Count', 'Hatespeech Count', 'Hatespeech Rat

print('\n--- Threshold comparison saved to 7_top5_hatespeech_targets_comparison.
print('--- Final top 5 results saved to 7_top5_hatespeech_targets.txt/png ---')

```

--- Ratio Top 5 (min_count = 30) ---

Target Type	Target	Total Count	Hatespeech Count	Hatespeech Ratio (%)
Religion	Jewish	1950	1406	72.102564
Race	Hispanic	313	213	68.051118
Race	African	3769	2513	66.675511
Race	Arab	1172	755	64.419795
Race	Indigenous	42	25	59.523810

--- Ratio Top 5 (min_count = 50) ---

Target Type	Target	Total Count	Hatespeech Count	Hatespeech Ratio (%)
Religion	Jewish	1950	1406	72.102564
Race	Hispanic	313	213	68.051118
Race	African	3769	2513	66.675511
Race	Arab	1172	755	64.419795
Religion	Islam	2559	1497	58.499414

--- Ratio Top 5 (min_count = 100) ---

Target Type	Target	Total Count	Hatespeech Count	Hatespeech Ratio (%)
Religion	Jewish	1950	1406	72.102564
Race	Hispanic	313	213	68.051118
Race	African	3769	2513	66.675511
Race	Arab	1172	755	64.419795
Religion	Islam	2559	1497	58.499414

Recommended minimum count threshold: 100

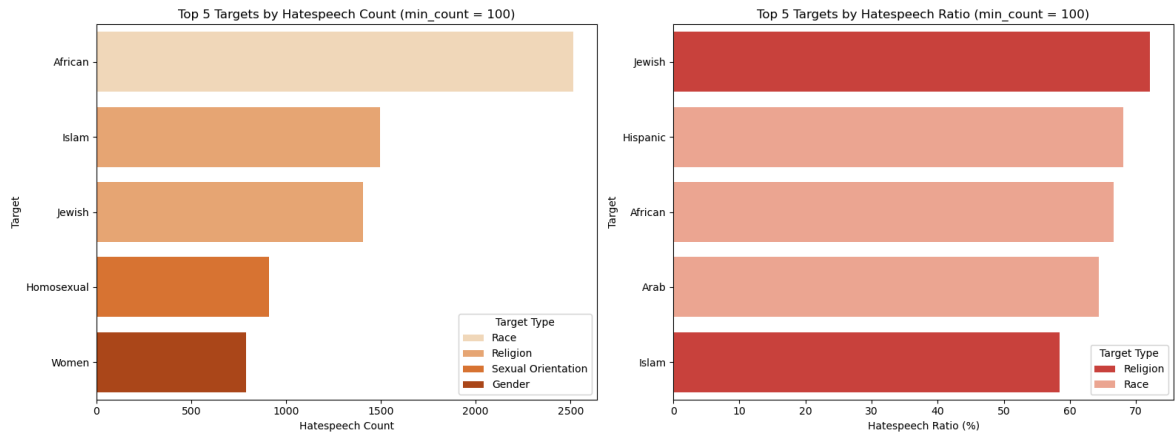
Reason: if we want groups that are actually attacked often, a larger total-count cutoff is safer.

--- Top 5 targets by hatespeech count ---

	Target Type	Target	Total Count	Hatespeech Count	Hatespeech Ratio (%)
1	Race	African	3769	2513	66.67551
4	Religion	Islam	2559	1497	58.49941
4	Religion	Jewish	1950	1406	72.10256
9	Sexual Orientation	Homosexual	2163	911	42.11742
5	Gender	Women	2262	790	34.92484

--- Top 5 targets by hatespeech ratio ---

Target Type	Target	Total Count	Hatespeech Count	Hatespeech Ratio (%)
Religion	Jewish	1950	1406	72.102564
Race	Hispanic	313	213	68.051118
Race	African	3769	2513	66.675511
Race	Arab	1172	755	64.419795
Religion	Islam	2559	1497	58.499414



--- Miscellaneous targets summary (separate) ---

label	Total Count	Hatespeech Count	Hatespeech Ratio (%)
Miscellaneous			
Other	2017	624	30.937035
Refugee	1274	455	35.714286
Minority	107	82	76.635514

--- Threshold comparison saved to 7_top5_hatespeech_targets_comparison.txt ---
 --- Final top 5 results saved to 7_top5_hatespeech_targets.txt/png ---

The top 5 targets with highest hate speech were analyzed using both total count and ratio. Total count of hate speech can reflect the overall prevalence of discrimination, while the ratio can capture how strongly hate speech is concentrated within each specific target.

The results show a clear distinction between frequency and concentration of hate speech across different groups. Targets such as African and Islam appear among the highest in total count, indicating that they are frequently mentioned and repeatedly exposed to hate speech. In contrast, groups such as Jewish and Hispanic exhibit highest hate speech ratios, suggesting that when these groups are mentioned, the comments are more likely to be explicitly hateful rather than neutral or general. This distinction highlights two different patterns of online discrimination: some groups experience widespread exposure to hate speech, while others are less frequently mentioned but are more likely to be targeted with direct and explicit hostility when they are referenced. In addition, African and Jewish groups are both included in the top 5 count and ratio. This indicates that both groups experience more frequent hate speech online. The distribution also shows that the religion and the race was the only two groups appeared in the top 5 highest hate speech ratio, reflecting that those two groups are the most visible and frequently discriminated online.

The miscellaneous group consists of heterogeneous features that do not belong to a single, coherent demographic dimension. Rather than forming a unified group, these features represent different underlying dimensions that are not explicitly structured in the dataset. For example, minority may relate to broader demographic vulnerability, economic reflects socioeconomic status, disability corresponds to health or ability status, and refugee relates to legal or citizenship status. Because these features belong to different conceptual axes, grouping them together under a single category would obscure their distinct meanings. Therefore, they were not included in the group-level comparisons and were instead treated separately in the analysis.

To ensure the stability of hate speech ratios, the minimum count threshold was set as 100. Multiple minimum count thresholds (30, 50, and 100) were tested. Lower thresholds include smaller groups that can reflect a double-marginalization, but higher thresholds were used to focus on more general and reliable estimates.

8. Top 5 Targets with Highest Offensive Count and Ratio

```
In [9]: # 8. Top 5 targets by offensive count and ratio
analysis_df = df.copy()
target_label = 'offensive'

# treat "no target" style values as missing targets for ranking
no_target_map = {
    'Race': 'No_race',
    'Religion': 'Nonreligious',
    'Gender': 'No_gender',
    'Sexual Orientation': 'No_orientation',
}

main_target_cols = ['Race', 'Religion', 'Gender', 'Sexual Orientation']
target_rows = []

for col in main_target_cols:
    valid_mask = analysis_df[col].notna() & (analysis_df[col] != no_target_map[col])
    col_df = analysis_df.loc[valid_mask, [col, 'label']].copy()

    for target_name, group in col_df.groupby(col):
        total_count = len(group)
        target_label_count = (group['label'] == target_label).sum()
        target_label_ratio = (target_label_count / total_count) * 100

        target_rows.append({
            'Target Type': col,
            'Target': target_name,
            'Total Count': total_count,
            'Offensive Count': target_label_count,
            'Offensive Ratio (%)': target_label_ratio,
        })

target_df = pd.DataFrame(target_rows)

# compare thresholds first, but use 100 for the final rankings
threshold_options = [30, 50, 100]

with open('8_top5_offensive_targets_comparison.txt', 'w') as f:
    f.write('--- Threshold Comparison for Ratio-Based Top 5 (excluding Miscellaneous) ---\n')

    for min_count in threshold_options:
        threshold_df = target_df[target_df['Total Count'] >= min_count].copy()
        ratio_top5 = threshold_df.sort_values(
            by=['Offensive Ratio (%)', 'Offensive Count', 'Total Count'],
            ascending=[False, False, False],
        ).head(5)

        print(f'--- Offensive Ratio Top 5 (min_count = {min_count}) ---')
```

```

print(ratio_top5.to_string(index=False))
print()

f.write(f'--- min_count = {min_count} ---\n')
f.write(ratio_top5.to_string(index=False))
f.write('\n\n')

recommended_min_count = 100
recommended_df = target_df[target_df['Total Count'] >= recommended_min_count].co

top5_by_count = recommended_df.sort_values(
    by=['Offensive Count', 'Offensive Ratio (%)', 'Total Count'],
    ascending=[False, False, False],
).head(5)

top5_by_ratio = recommended_df.sort_values(
    by=['Offensive Ratio (%)', 'Offensive Count', 'Total Count'],
    ascending=[False, False, False],
).head(5)

print(f'Recommended minimum count threshold: {recommended_min_count}')
print('Reason: if we want groups that are actually attacked often, a larger tota
print('\n--- Top 5 targets by offensive count ---')
print(top5_by_count.to_string(index=False))
print('\n--- Top 5 targets by offensive ratio ---')
print(top5_by_ratio.to_string(index=False))

with open('8_top5_offensive_targets.txt', 'w') as f:
    f.write('--- Final Top 5 Offensive Targets (excluding Miscellaneous) ---\n')
    f.write(f'Minimum total count threshold: {recommended_min_count}\n')
    f.write('Reason: use 100+ observations so the top 5 reflects groups with mea
    f.write('--- Top 5 by Offensive Count ---\n')
    f.write(top5_by_count.to_string(index=False))
    f.write('\n\n--- Top 5 by Offensive Ratio (%) ---\n')
    f.write(top5_by_ratio.to_string(index=False))
    f.write('\n\n--- Full Target Table at Recommended Threshold ---\n')
    f.write(recommended_df.sort_values(
        by=['Offensive Count', 'Offensive Ratio (%)', 'Total Count'],
        ascending=[False, False, False],
    ).to_string(index=False))

fig, axes = plt.subplots(1, 2, figsize=(16, 6))

sns.barplot(
    data=top5_by_count,
    x='Offensive Count',
    y='Target',
    hue='Target Type',
    dodge=False,
    palette='Blues',
    ax=axes[0],
)
axes[0].set_title(f'Top 5 Targets by Offensive Count (min_count = {recommended_m
axes[0].set_xlabel('Offensive Count')
axes[0].set_ylabel('Target')
axes[0].legend(title='Target Type', loc='lower right')

sns.barplot(
    data=top5_by_ratio,
    x='Offensive Ratio (%)',

```

```

    y='Target',
    hue='Target Type',
    dodge=False,
    palette='PuBu',
    ax=axes[1],
)
axes[1].set_title(f'Top 5 Targets by Offensive Ratio (min_count = {recommended_min_count})')
axes[1].set_xlabel('Offensive Ratio (%)')
axes[1].set_ylabel('Target')
axes[1].legend(title='Target Type', loc='lower right')

plt.tight_layout()
plt.savefig('9_top5_offensive_targets.png', bbox_inches='tight', dpi=300)
plt.show()

# optional side analysis: Miscellaneous only
misc_df = analysis_df[analysis_df['Miscellaneous'].notna() & (analysis_df['Miscellaneous'] != 'None')]

if not misc_df.empty:
    misc_summary = misc_df.groupby('Miscellaneous')['label'].value_counts().unstack()
    misc_summary['Total Count'] = misc_summary.sum(axis=1)
    misc_summary['Offensive Count'] = misc_summary.get(target_label, 0)
    misc_summary['Offensive Ratio (%)'] = (
        misc_summary['Offensive Count'] / misc_summary['Total Count'] * 100
    )
    misc_summary = misc_summary[misc_summary['Total Count'] >= recommended_min_count]
    by=['Offensive Count', 'Offensive Ratio (%)', 'Total Count'],
    ascending=[False, False, False],
)

print('\n--- Miscellaneous offensive targets summary (separate) ---')
print(misc_summary[['Total Count', 'Offensive Count', 'Offensive Ratio (%)']])

with open('8_miscellaneous_offensive_targets.txt', 'w') as f:
    f.write('--- Miscellaneous Offensive Target Summary ---\n')
    f.write(f'Minimum total count threshold: {recommended_min_count}\n\n')
    f.write(misc_summary[['Total Count', 'Offensive Count', 'Offensive Ratio (%)']])

print('\n--- Threshold comparison saved to 8_top5_offensive_targets_comparison.txt ---')
print('--- Final top 5 results saved to 8_top5_offensive_targets.txt/png ---')

```

--- Offensive Ratio Top 5 (min_count = 30) ---

	Target Type	Target	Total Count	Offensive Count	Offensive Ratio (%)
4	Gender	Women	2262	1158	51.19363
3	Race	Caucasian	802	399	49.75062
0	Race	Indian	32	14	43.75000
4	Sexual Orientation	Homosexual	2163	913	42.20989
9	Sexual Orientation	Heterosexual	113	45	39.82300

--- Offensive Ratio Top 5 (min_count = 50) ---

	Target Type	Target	Total Count	Offensive Count	Offensive Ratio (%)
4	Gender	Women	2262	1158	51.19363
3	Race	Caucasian	802	399	49.75062
4	Sexual Orientation	Homosexual	2163	913	42.20989
9	Sexual Orientation	Heterosexual	113	45	39.82300
0	Race	Asian	400	145	36.25000

--- Offensive Ratio Top 5 (min_count = 100) ---

	Target Type	Target	Total Count	Offensive Count	Offensive Ratio (%)
4	Gender	Women	2262	1158	51.19363
3	Race	Caucasian	802	399	49.75062
4	Sexual Orientation	Homosexual	2163	913	42.20989
9	Sexual Orientation	Heterosexual	113	45	39.82300
0	Race	Asian	400	145	36.25000

Recommended minimum count threshold: 100

Reason: if we want groups that are actually attacked often, a larger total-count cutoff is safer.

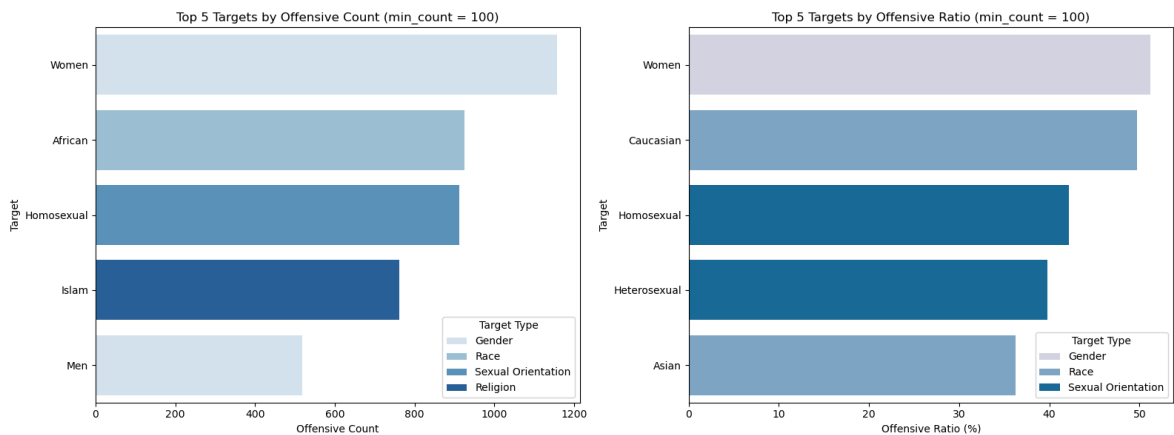
--- Top 5 targets by offensive count ---

	Target Type	Target	Total Count	Offensive Count	Offensive Ratio (%)
	Gender	Women	2262	1158	51.193634
	Race	African	3769	926	24.568851
	Sexual Orientation	Homosexual	2163	913	42.209894
	Religion	Islam	2559	761	29.738179
	Gender	Men	1467	518	35.310157

--- Top 5 targets by offensive ratio ---

	Target Type	Target	Total Count	Offensive Count	Offensive Ratio (%)
4	Gender	Women	2262	1158	51.19363
	Race	Caucasian	802	399	49.75062

3	Sexual Orientation	Homosexual	2163	913	42.20989
4	Sexual Orientation	Heterosexual	113	45	39.82300
9	Race	Asian	400	145	36.25000
0					



```

--- Miscellaneous offensive targets summary (separate) ---
label          Total Count  Offensive Count  Offensive Ratio (%)
Miscellaneous
Other           2017        1029             51.016361
Refugee         1274         504             39.560440
Minority        107          25              23.364486

```

```

--- Threshold comparison saved to 8_top5_offensive_targets_comparison.txt ---
--- Final top 5 results saved to 8_top5_offensive_targets.txt/png ---

```

The analysis of the top 5 targets based on offensive language count and ratio reveals a different pattern compared to hate speech, highlighting variation in how hostility is expressed across groups.

Groups such as women, African, and homosexual individuals rank among the highest in total offensive count, indicating that these groups are frequently mentioned in contexts involving offensive language. This suggests a high level of exposure to non-explicit but negative or derogatory expressions. In contrast, the highest offensive comment ratios are observed for women and Caucasian targets, followed by homosexual and heterosexual subgroups. This indicates that when these groups are mentioned, a substantial proportion of comments contain offensive language, though not necessarily explicit hate speech.

Notably, the subgroup women ranks highest both offensive count and ratio, and it is also the only subgroup where offensive comments exceed hate speech comment. This suggests that hostility toward women is more often expressed through normalized or indirect forms of negative language rather than direct speech. These patterns indicate that offensive comments capture a different dimension of online hostility compared to hate speech. While hate speech tends to be more concentrated for race and religious groups, offensive comments appear more widespread, particularly for gender-related and sexuality-related groups.

To ensure the stability of these patterns, the same minimum count thresholds to the top 5 hate speech analysis were tested and applied. The composition of the top

groups remained largely consistent across thresholds, with women, Caucasian, and homosexual groups consistently appearing among those with the highest offensive ratios. The miscellaneous group was also treated in the same way to the top 5 hate speech analysis. Within the group, the subgroup, other, show relatively high offensive ratios, while other subgroups such as minority show lower proportions, further illustrating that offensive comment patterns vary depending on the underlying conceptual dimension.

9. Top 5 Targets with Highest Normal Count and Ratio

```
In [10]: # 9. Top 5 targets by normal count and ratio
analysis_df = df.copy()
target_label = 'normal'

# treat "no target" style values as missing targets for ranking
no_target_map = {
    'Race': 'No_race',
    'Religion': 'Nonreligious',
    'Gender': 'No_gender',
    'Sexual Orientation': 'No_orientation',
}

main_target_cols = ['Race', 'Religion', 'Gender', 'Sexual Orientation']
target_rows = []

for col in main_target_cols:
    valid_mask = analysis_df[col].notna() & (analysis_df[col] != no_target_map[col])
    col_df = analysis_df.loc[valid_mask, [col, 'label']].copy()

    for target_name, group in col_df.groupby(col):
        total_count = len(group)
        target_label_count = (group['label'] == target_label).sum()
        target_label_ratio = (target_label_count / total_count) * 100

        target_rows.append({
            'Target Type': col,
            'Target': target_name,
            'Total Count': total_count,
            'Normal Count': target_label_count,
            'Normal Ratio (%)': target_label_ratio,
        })

target_df = pd.DataFrame(target_rows)

# compare thresholds first, but use 100 for the final rankings
threshold_options = [30, 50, 100]

with open('9_top5_normal_targets_comparison.txt', 'w') as f:
    f.write('--- Threshold Comparison for Ratio-Based Top 5 (excluding Miscellaneous)\n')

    for min_count in threshold_options:
        threshold_df = target_df[target_df['Total Count'] >= min_count].copy()
        ratio_top5 = threshold_df.sort_values(
            by=['Normal Ratio (%)', 'Normal Count', 'Total Count'],
```

```

        ascending=[False, False, False],
    ).head(5)

    print(f'--- Normal Ratio Top 5 (min_count = {min_count}) ---')
    print(ratio_top5.to_string(index=False))
    print()

    f.write(f'--- min_count = {min_count} ---\n')
    f.write(ratio_top5.to_string(index=False))
    f.write('\n\n')

recommended_min_count = 100
recommended_df = target_df[target_df['Total Count'] >= recommended_min_count].co

top5_by_count = recommended_df.sort_values(
    by=['Normal Count', 'Normal Ratio (%)', 'Total Count'],
    ascending=[False, False, False],
).head(5)

top5_by_ratio = recommended_df.sort_values(
    by=['Normal Ratio (%)', 'Normal Count', 'Total Count'],
    ascending=[False, False, False],
).head(5)

print(f'Recommended minimum count threshold: {recommended_min_count}')
print('Reason: if we want groups that are actually mentioned often, a larger tot
print('\n--- Top 5 targets by normal count ---')
print(top5_by_count.to_string(index=False))
print('\n--- Top 5 targets by normal ratio ---')
print(top5_by_ratio.to_string(index=False))

with open('9_top5_normal_targets.txt', 'w') as f:
    f.write('--- Final Top 5 Normal Targets (excluding Miscellaneous) ---\n')
    f.write(f'Minimum total count threshold: {recommended_min_count}\n')
    f.write('Reason: use 100+ observations so the top 5 reflects groups with mea
    f.write('--- Top 5 by Normal Count ---\n')
    f.write(top5_by_count.to_string(index=False))
    f.write('\n\n--- Top 5 by Normal Ratio (%) ---\n')
    f.write(top5_by_ratio.to_string(index=False))
    f.write('\n\n--- Full Target Table at Recommended Threshold ---\n')
    f.write(recommended_df.sort_values(
        by=['Normal Count', 'Normal Ratio (%)', 'Total Count'],
        ascending=[False, False, False],
    ).to_string(index=False))

fig, axes = plt.subplots(1, 2, figsize=(16, 6))

sns.barplot(
    data=top5_by_count,
    x='Normal Count',
    y='Target',
    hue='Target Type',
    dodge=False,
    palette='Greens',
    ax=axes[0],
)
axes[0].set_title(f'Top 5 Targets by Normal Count (min_count = {recommended_min_
axes[0].set_xlabel('Normal Count')
axes[0].set_ylabel('Target')
axes[0].legend(title='Target Type', loc='lower right')

```

```

sns.barplot(
    data=top5_by_ratio,
    x='Normal Ratio (%)',
    y='Target',
    hue='Target Type',
    dodge=False,
    palette='YlGn',
    ax=axes[1],
)
axes[1].set_title(f'Top 5 Targets by Normal Ratio (min_count = {recommended_min_
axes[1].set_xlabel('Normal Ratio (%)')
axes[1].set_ylabel('Target')
axes[1].legend(title='Target Type', loc='lower right')

plt.tight_layout()
plt.savefig('9_top5_normal_targets.png', bbox_inches='tight', dpi=300)
plt.show()

# optional side analysis: Miscellaneous only
misc_df = analysis_df[analysis_df['Miscellaneous'].notna() & (analysis_df['Misce

if not misc_df.empty:
    misc_summary = misc_df.groupby('Miscellaneous')['label'].value_counts().unst
    misc_summary['Total Count'] = misc_summary.sum(axis=1)
    misc_summary['Normal Count'] = misc_summary.get(target_label, 0)
    misc_summary['Normal Ratio (%)'] = (
        misc_summary['Normal Count'] / misc_summary['Total Count'] * 100
    )
    misc_summary = misc_summary[misc_summary['Total Count'] >= recommended_min_c
        by=['Normal Count', 'Normal Ratio (%)', 'Total Count'],
        ascending=[False, False, False],
    )

    print('\n--- Miscellaneous normal targets summary (separate) ---')
    print(misc_summary[['Total Count', 'Normal Count', 'Normal Ratio (%)']].to_s

    with open('9_miscellaneous_normal_targets.txt', 'w') as f:
        f.write('--- Miscellaneous Normal Target Summary ---\n')
        f.write(f'Minimum total count threshold: {recommended_min_count}\n\n')
        f.write(misc_summary[['Total Count', 'Normal Count', 'Normal Ratio (%)']

print('\n--- Threshold comparison saved to 9_top5_normal_targets_comparison.txt
print('--- Final top 5 results saved to 9_top5_normal_targets.txt/png ---')

```

--- Normal Ratio Top 5 (min_count = 30) ---

Target Type	Target	Total Count	Normal Count	Normal Ratio (%)
Gender	Men	1467	471	32.106339
Religion	Christian	163	51	31.288344
Religion	Hindu	42	11	26.190476
Race	Caucasian	802	199	24.812968
Race	Asian	400	92	23.000000

--- Normal Ratio Top 5 (min_count = 50) ---

Target Type	Target	Total Count	Normal Count	Normal Ratio (%)
Gender	Men	1467	471	32.106339
Religion	Christian	163	51	31.288344
Race	Caucasian	802	199	24.812968
Race	Asian	400	92	23.000000
Sexual Orientation	Homosexual	2163	339	15.672677

--- Normal Ratio Top 5 (min_count = 100) ---

Target Type	Target	Total Count	Normal Count	Normal Ratio (%)
Gender	Men	1467	471	32.106339
Religion	Christian	163	51	31.288344
Race	Caucasian	802	199	24.812968
Race	Asian	400	92	23.000000
Sexual Orientation	Homosexual	2163	339	15.672677

Recommended minimum count threshold: 100

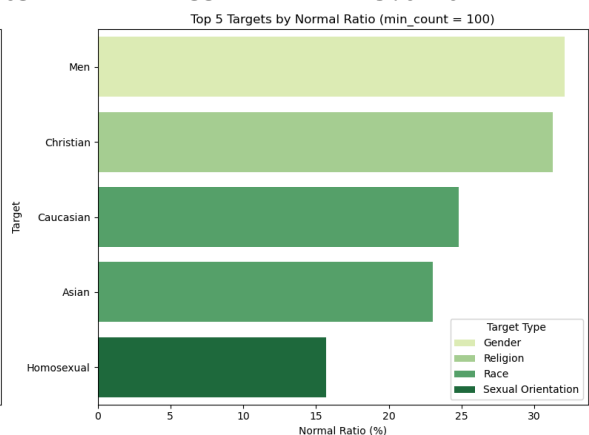
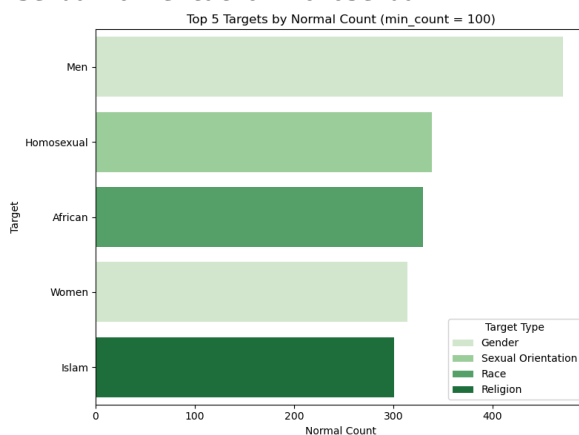
Reason: if we want groups that are actually mentioned often, a larger total-count cutoff is safer.

--- Top 5 targets by normal count ---

Target Type	Target	Total Count	Normal Count	Normal Ratio (%)
Gender	Men	1467	471	32.106339
Sexual Orientation	Homosexual	2163	339	15.672677
Race	African	3769	330	8.755638
Gender	Women	2262	314	13.881521
Religion	Islam	2559	301	11.762407

--- Top 5 targets by normal ratio ---

Target Type	Target	Total Count	Normal Count	Normal Ratio (%)
Gender	Men	1467	471	32.106339
Religion	Christian	163	51	31.288344
Race	Caucasian	802	199	24.812968
Race	Asian	400	92	23.000000
Sexual Orientation	Homosexual	2163	339	15.672677



```

--- Miscellaneous normal targets summary (separate) ---
label          Total Count  Normal Count  Normal Ratio (%)
Miscellaneous
Other           2017         364          18.046604
Refugee         1274         315          24.725275
Minority        107           0           0.000000

```

```

--- Threshold comparison saved to 9_top5_normal_targets_comparison.txt ---
--- Final top 5 results saved to 9_top5_normal_targets.txt/png ---

```

The analysis of the top 5 targest based on normal comment count and ratio provides a complementray perspective to hate speech and offensive comments, highlighting where discussions are more likely to remain neutral.

Groups such as men, homosexual individuals, and African appear among the highest in total normal count,indicating that these groups are frequently mentioned in general or non-hostile contexts. This suggests relatively higher visibility in broader discussions beyond explicitly negative lanaaguage. In contract, the highest normal comment ratios are observed for men and Christian groups, followed by Caucasian and Asian groups. This indicates that when these groups are mentioned, the comments are more likely to be neutral rather than offensive or hateful.

In English-speaking western society, both men and Christian groups are considered as majority, so the men group ranked the top 1 may not be surprising. What is significant here is African, homosexual, women, and Islam groups in the top 5 targets by normal comment count. They appeared in the top 5 groups that received the most hate speech or offensive comments and in total count or ratio. This may seem contradictory, but the previous results should be considered together. When reviewing the speech label distribution for each of these groups, women group received offensive comments the most (51.2%) and followed by hate speech (34.9%), Caucasian received the offensive the most (49.8%) and followed by hate speech (25.4%) which is very close to normal speech rate (24.8%). Otherwise, the groups in the top 5 normal comment list except men group actually received hate speech comment the most and then offensive comment the next. These findings highlight that high frequency in normal comments does not necessarily indicate a neutral representation. Instead, it often reflects high overall mention rates, where both neutral and negative discourse coexist. Therefore, interpreting these results requires considering how the same groups appear across multiple categories rather than viewing normal comment counts in isolation.

III. Results and Discussion

The results show that online comment surrounding demographic groups is structured not only by how frequently groups are mentioned, but also by how language is directed toward them. Across the dataset, comments that explicitly target a group are more likely to contain offensive language or hate speech rather than remain neutral. This indicates that the act of targeting itself is associated with a shift toward non-neutral language. At the same time, differences across groups reveal that similar

levels of visibility do not lead to similar patterns of representation. Groups such as African, women, homosexual, and Islam are frequently mentioned but are more likely to be associated with offensive or hateful language. In contrast, groups such as men and Christian maintain relatively higher proportions of normal comments, suggesting that mentions of these groups are more likely to remain neutral.

Three key patterns emerge from the analysis of label distributions and top target groups. First, comments that explicitly target a demographic group are more likely to contain offensive language or hate speech rather than remain neutral. This suggests that once a target is referenced, the probability of non-neutral language increases, indicating that targeting itself is closely associated with negativity. Second, despite this overall pattern, a notable exception appears. The men subgroup stands out as the only group in which normal comments are the most prevalent in terms of total count and ratio. This indicates that not all highly visible groups are associated with negative language to the same extent, and that certain groups are more likely to be discussed in neutral contexts. Third, comparing different analytical perspectives reveals that interpretations depend strongly on the chosen metric. Total counts emphasize exposure, while ratios highlight concentration of hostility. As a result, the same dataset can produce different conclusions depending on whether the analysis focuses on frequency or proportional intensity.

Taken together, these findings indicate that patterns of online discourse are shaped not only by how often groups are mentioned, but also by how those mentions are framed. Understanding these patterns requires considering both visibility and the direction of language, rather than relying on a single measure

IV. Conclusion

This study shows that online language directed toward demographic groups is not neutral by default. Once a group is explicitly referenced, the likelihood of offensive or hateful language increases.

However, interpreting these patterns requires more than examining frequency and ratio alone. Differences in how groups are represented cannot be fully understood without considering background context, such as the proportion of each demographic group within the overall population. For example, groups such as men and women represent a large share of the population, which may naturally lead to higher visibility in the dataset. In addition, platform-specific factors - such as the demographic composition of users and the cultural or communicative norms of the social media environment - may influence how language is expressed and directed toward different groups.

Therefore, while this analysis identifies clear patterns in the distribution and direction of language, a more complete understanding requires integrating these findings with contextual information beyond the dataset itself. This highlights the importance of

combining quantitative patterns with broader social context when interpreting online discourse.